Short-lead forecasting of GRACE terrestrial water storage is a noise-filtering problem

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Abstract. Forecasts of GRACE and GRACE-FO terrestrial water storage anomalies (TWSA) are routinely benchmarked against persistence. Persistence of a noisy observation, however, carries the anchor month's measurement noise into every forecast. We propose a stronger member of the same reference class: persistence of the filtered state, obtained from a three-parameter AR(1)-plus-observation-noise model fit per basin. On 234 basins worldwide over five expanding-window test folds, this filter beats damped persistence by 5.0 and 8.8 % at leads of one and two months and per-basin ridge regression at five of six leads. Refitting the same model with the noise variance fixed at zero removes the entire margin, so the gain is the noise filtering itself, not the autoregressive structure – and much of the published skill of learned models over persistence is attainable by filtering alone. What beats the filter is exogenous forcing: ERA5-Land predictors add 4.6 % at lead one, and a twelve-month forcing history extends the gain through lead three – read linearly, since a ridge over the flattened history outperforms every sequence model we trained on it. A head-to-head comparison with the forcing-driven GRACE-FCast system of Li and Kusche (2026) on 227 matched basins locates the initial-conditions-versus-forcing crossover on satellite-observed TWSA: our system wins at lead one by 20 %, the two are not detectably different at lead two, and theirs wins from lead three. Neighboring basins add nothing detectable to linear models, and the residual linear effect concentrates in basins whose mascon footprints are shared with land outside the basin – a leakage signature. The same neighbor signal, propagated to the forecast target time and applied as a residual correction to a deep sequence model, adds 0.9–2.0 % at every lead, beats every degree-matched random-graph placebo, and shows none of the leakage signature; given the neighbor's raw history instead, the identical correction stage recovers nothing. Short-lead TWSA forecasting is a filtering problem; beyond two to three months it becomes a forcing problem, and the two skill sources are complementary rather than competing.

1 Introduction

The GRACE and GRACE-FO satellite missions have measured monthly changes in terrestrial water storage (TWS) since 2002, integrating groundwater, soil moisture, surface water, snow, and ice into a single observable that no in situ network can replicate (Tapley et al., 2004; Landerer et al., 2020). Anomalies of this quantity (TWSA) carry months of memory, which makes them both a target and a source of predictability: TWSA forecasts support drought early warning and water management (Li et al., 2026; Arsenault et al., 2020), flood-potential assessment (Reager et al., 2014), and the bridging of the 2–3 month latency of GRACE-FO data releases (Mo et al., 2025).

A growing literature forecasts TWSA directly from data. Neural networks have been trained on GRACE and hydrometeorological series at basin scale (Ahmed et al., 2019), at grid scale globally (Li et al., 2020, 2024; Li and Kusche, 2026), for Europe with subsequent assimilation (Li et al., 2025), and for latency-filling (Mo et al., 2025); statistical time-series methods have been applied per basin (Ahi and Cekim, 2021); benchmark predictability has been quantified at decadal scale (Zhu et al., 2019); and physically based systems forecast TWS with seasonal meteorological forcing (Li et al., 2026; Getirana et al., 2020). Nearly all of these studies quantify their skill against some form of persistence or climatology, and the implicit logic is that whatever beats persistence has extracted predictive structure beyond simple memory.

This paper starts from an observation that undermines that logic for GRACE specifically. Basin-average TWSA is not a state variable observed cleanly; it is a state variable observed through a noisy instrument, with observation errors that are large for small and arid basins (Save et al., 2016; Landerer et al., 2020). A persistence forecast – last month's observed anomaly, damped or not – propagates the observation noise of the anchor month into every forecast. The proper persistence-class reference for a noisy observable is not persistence of the observation but persistence of the *filtered state*: an estimate of the signal with the observation noise removed. For an AR(1) signal observed with white noise this optimal filter is the Kalman filter (Kalman, 1960), and it costs three parameters per basin. If this stronger member of the persistence class beats standard persistence by several percent, then part of the reported skill of learned models over persistence is not learned structure at all – it is noise filtering that a two-line state-space model performs for free.

The same benchmark-hygiene argument has been made in sea-ice forecasting, where Niraula and Goessling (2021) showed that damped-persistence references that "do not exploit observations optimally" flatter dynamical forecast systems, and built a stronger persistence-class benchmark from spatial damping. We execute the analogous move for satellite-observed TWSA via explicit signal/observation-noise separation. Within the GRACE literature, Kalman machinery is well established but serves different purposes: ensemble Kalman filters assimilate TWSA into land-surface models (Springer et al., 2026), and state-space smoothers densify or clean the gravity-field record itself (Kankanige et al., 2026). Neither uses a univariate AR(1)-plus-observation-noise model on the finished basin series *as a forecast benchmark*, which is the role it plays here; our filter is not a data-assimilation system and updates no model states. Related in spirit, Ahi and Cekim (2021) applied exponential-smoothing family methods per basin without a noise-separation argument, and Kankanige et al. (2026) found persistence-based prediction of GRACE spherical-harmonic coefficients competitive at sub-monthly scales – independent support that the persistence class is underrated. Most directly, Nie et al. (2025) showed on 515 basins that deep architectures do not beat linear regression for TWS prediction when autoregressive information is deliberately excluded from the feature set; we show the complementary result that when autoregressive information is *included*, how one filters it dominates the model comparison.

If short-lead skill is mostly filtering, what is left for models to earn? The seasonal hydrologic forecasting literature answered a structurally identical question long ago: skill at short leads comes from initial conditions, and skill at long leads from meteorological forcing, with a crossover in between (Wood and Lettenmaier, 2008; Shukla and Lettenmaier, 2011; Koster et al., 2010; Yossef et al., 2013; Pechlivanidis et al., 2020). That partition was established with physically based models in ESP/reverse-ESP experiments, mostly for streamflow and soil moisture. It has not previously been located on satellite-observed TWSA, nor measured between two real, published, data-driven forecast systems. We do both, comparing our observation-noise-aware system head-to-head with the GRACE-FCast hindcasts of Li and Kusche (2026), whose LSTM ingests exogenous hydrometeorological predictors and, by design, no GRACE lags. The comparison locates an empirical crossing at approximately two months. We describe it as consistent with the initial-conditions/forcing partition rather than as a decomposition of skill into sources: the two systems differ along many axes at once, so what is measured is the lead profile, not the effect of any single design choice.

The third question the paper answers is whether TWSA fields themselves carry usable cross-basin information – whether one basin's history helps forecast another's future once own-basin filtering and local meteorology are accounted for. Spatial and graph-based TWSA methods increasingly assume so: Li and Kusche (2026) shift predictors across space, Arzoumanidis et al. (2026) build a proximity-plus-correlation basin graph for TWSA reconstruction, and Steidl and Zhu (2025) forecast TWSA with a hierarchical basin-graph network. None of these works isolates the incremental contribution of the cross-basin information, and none tests the graph against a null; Steidl and Zhu (2025) additionally target a reconstructed rather than observed TWSA product. To our knowledge, no prior work measures the incremental cross-basin information in *observed* GRACE TWSA against null models. We do so with degree-matched random graphs, a mascon leakage distance exclusion, a resolution stratification, and full ERA5 local-meteorology conditioning. The answer is two-sided, and the two sides must be kept apart: at the linear tier the incremental effect is a null, while the same neighbor signal does deliver skill through a dedicated nonlinear correction stage. We therefore also show what governs whether the signal is usable by a deep model at all, and it is not the architecture but the *representation*: the neighbor helps only when it arrives as a state already propagated to the forecast target time, and is inert as raw history however it is delivered.

The contributions, in order:

1. **An observation-noise-aware benchmark.** A per-basin three-parameter Kalman filter (AR(1) signal plus observation noise) beats damped persistence by 5.0–8.8 % at leads 1–2 and per-basin ridge regression at five of six leads (DM $p \leq 0.013$; lead 4, $p = 0.078$, is not significant), making the filtering class the strongest own-basin reference at every lead. Benchmarked against persistence, a substantial share of published "ML skill" on deseasonalized TWSA is therefore reproduced by three-parameter state-space filtering alone. The interesting question becomes what *does* beat the filter; the answer – exogenous forcing, a longer window of that forcing read linearly, and a small spatial correction – maps where genuine information lives.

2. **The initial-conditions/forcing crossover, located on observed TWSA.** Against the published GRACE-FCast hindcast (Li and Kusche, 2026; Li, 2025) on 227 matched basins under a common protocol, our best system wins lead 1 by 20.0 % (DM $p = 3.4 \times 10^{-4}$), is not detectably different at lead 2, and loses leads 3–6 by 13–30 %. The two systems are complementary, not competing: short leads are a filtering problem, long leads a forcing problem.

3. **Cross-basin information: a linear null, a nonlinear signal, and the representation that separates them.** Adding a selected neighbor to the linear benchmark changes nothing detectable, and the pooled lead-1 effect that survives concentrates in basins whose mascon footprints are shared with land outside the basin – a leakage signature, not hydrology (Sect. 4.4). Yet the same neighbor signal, propagated to the forecast target time by its own estimated dynamics and ap-

plied as a residual correction to a deep sequence model, adds $+0.9$ to $+2.0\,\%$ at every lead, beats every degree-matched random-graph placebo, and shows none of the leakage signature. The representation decides: fed the neighbor's raw 12-month history instead of the propagated state, the identical correction stage recovers nothing (Sect. 4.5).

Section 2 describes the data, Sect. 3 the decomposition, benchmark, controls, and learned models, Sect. 4 the results in the order above, and Sect. 5 the implications, the comparability of published numbers, and the limitations.

2 Data

2.1 GRACE/GRACE-FO mascons and basin sample

We use CSR RL0603M mascon solutions (Save et al., 2016) on a $0.25°$ grid, spanning April 2002 through May 2026, expressed as equivalent water height (cm); the distributed file is titled RL0603M and reports product version RL06.2. Each of the 257 monthly solutions is assigned to its official calendar month using the product's documented missing-month list, with the solution's data span asserted to overlap its assigned month. Midpoint binning mislabels the November 2011 and May 2015 solutions, whose arcs are centered on 31 October 2011 and 27 April 2015; under that convention two months hold two solutions each and November 2011 and May 2015 are silently dropped from the record. Basins are defined by a HydroSHEDS-based Level-3 basin–mascon partition (Lehner and Grill, 2013); each basin series is the cosine-latitude area-weighted mean of its mask cells, with longitude centroids computed by circular mean. From the 284 basins in the partition we exclude 46 ice-sheet basins (Antarctica, Greenland) and 4 water bodies, leaving **234 basins** on all inhabited continents; 38 of these are African (37 mainland plus Madagascar), and 8 are glaciated and tracked as a separate stratum.

2.1.1 Basin areas and native mascon geometry.

Basin areas used in the stratifications of Sect. 4.4 are our own cosine-latitude integration of the mask file. Native mascon assignment comes from CSR's official ancillary products: the RL06.3 native-mascon-ID mapping file, which labels every $0.25°$ grid cell with its native mascon, and the RL06 v02 land/ocean masks (doi:10.15781/cgq9-nh24; Save et al., 2016), which CSR recommends for basin definitions to minimize coastal leakage. As an internal check we also recovered the tiles empirically – the gridded product replicates one tile value across the $0.25°$ cells it covers, so fingerprinting eight months and grouping cells on exact series equality reconstructs the partition – and the two agree *exactly*: 42,107 tiles in both, cell-level purity 1.0 in both directions, adjusted Rand index 1.0, and per-basin footprint-sharing scores identical to four decimals. The official tiles have median area 12,123 km^2 (IQR 11,518–13,032), with cell counts rising from 16 to 24 with latitude – near-constant area at varying cell count, CSR's $\sim 1°$ equal-area geodesic grid. A 90,000 km^2 basin therefore spans roughly *seven* native mascons, not one; only 3 of the 234 basins span fewer than two. We accordingly do not describe that threshold as one resolution element. CSR separately advises caution for basins below approximately 200,000 km^2, a cut under which 35 of our 199 nominally resolved basins fall; Sect. 4.4 therefore repeats the footprint stratification at that threshold as a sensitivity.

2.2 ERA5-Land forcing

We use 11 ERA5-Land variables (Muñoz-Sabater et al., 2021) at $0.5°$ (server-side regridded from $0.1°$), January 2001 through May 2026: total precipitation, evaporation, total/surface/subsurface runoff, volumetric soil water in four layers, 2-m temperature, and snow depth water equivalent. Accumulated fluxes are converted from mean daily rates to monthly totals (cm month^{-1}), evaporation is sign-flipped so positive means moisture loss, temperature is converted to °C, and snow depth to cm water equivalent. Basin aggregation uses the same cosine-latitude weighted means as the TWSA aggregation, mapping each $0.25°$ mask cell to its nearest $0.5°$ ERA5 cell. The resulting table (284 basins $\times$ 305 months) is gapless. Because ERA5-Land is defined over land only, coastal and island basins aggregate their forcing from the land fraction of their mask: of the 234 analysis basins, 79 draw less than 90 % of their mask weight from valid ERA5 land cells and 13 less than 50 % (minimum 24.5 %). These basins' forcing features average a sliver of coast and are correspondingly noisier; no basin is excluded for this, and the per-basin ERA5 results (Sect. 4.2) should be read with it in mind.

2.3 Climate indices

Twelve monthly climate indices (ENSO, IOD, NAO, and others) from NOAA PSL are used in the conditioning controls (Sect. 3.4) and in the pooled ridge's climate-index variant (Table 2); AMM and PDO end before the test window and are dropped from that variant. No headline model takes index features.

2.4 The GRACE-FCast hindcast

For the head-to-head comparison we use the public GRACE-FCast hindcast of Li and Kusche (2026) in its CSR-forced member (released as "CSR-FCast"), downloaded from PANGAEA (Li, 2025): monthly $1°$ global forecasts at leads 1–12 months, initialized December 2009 through May 2024 (174 initializations). Their LSTM forecasts TWSA from antecedent hydrometeorological and oceanographic predictors with correlation-based spatial predictor shifting; by design it takes no GRACE lags as input (Li and Kusche, 2026). We evaluate both their full-signal product and their non-seasonal variant (Sect. 3.6).

Two features of that choice bear on how our comparison should be read. GRACE-FCast is released as three products, built on the JPL, CSR and GSFC mascon solutions, and the skill figures reported in Li and Kusche (2026) are for the average of the three; the CSR member we use is the one their own training-residual diagnostic ranks least certain (3.0 versus 2.8 cm). We nonetheless compare against the CSR member alone because our own system is built on CSR mascons, so this is the mascon-matched contrast and the only one in which the two systems share a gravity solution — but it means our numbers characterize that member, not the product their headline scores describe. Second, their decomposition removes a linear trend as well as the seasonal cycle, so their non-seasonal variant and our deseasonalized target are defined on the same footing (Sect. 3.1); the functional form of their seasonal and trend fit follows Li et al. (2020) and is not restated in the forecast paper, which is why we also report the full-signal variant, where any such convention difference is absorbed by subtracting our own climatology.

3 Methods

3.1 Target definition: fold-safe deseasonalization

For each basin and each evaluation fold we fit, on the training window only, a climatology consisting of a linear trend plus annual and semiannual harmonics (six coefficients, ordinary least squares). The frozen coefficients deseasonalize both training and test data; the residual is divided by its training-window standard deviation. The forecast target is this deseasonalized, per-basin-standardized residual – the interannual anomalies that drought and flood applications need – and all pooled metrics are computed in this standardized space so that high-variance basins do not dominate. Physical-unit (cm) results are reported where noted.

3.2 Evaluation protocol

3.2.1 Folds.

Five expanding-window folds whose issue windows jointly cover June 2019 through April 2026 at lead 1, with targets reaching May 2026 (17, 17, 17, 17, and 15 issue months per basin at lead 1, one fewer per basin at each successive lead as the shifted target runs off the end of the record). The matched sample is therefore $n = 19{,}422/19{,}188/18{,}954/18{,}720/18{,}486/18{,}252$ rows at leads 1–6, identical for every model at a given lead. Fold membership is defined on forecast *issue* dates: within each fold, every climatology, filter parameter, model coefficient, neighbor graph, and standardization moment is fit strictly on observations before the fold's first issue date, and every test forecast is issued at or after that date – the model is frozen once and then used forward, as it would be in deployment. Splitting on target dates instead would let lead-2–6 transforms see up to $h-1$ months past the issue date, which is why the issue-date convention matters. Earlier issue windows are unusable: the 2017–2018 gap between GRACE and GRACE-FO destroys the lag windows needed to forecast across them. All results are retrospective potential skill: issue time is defined relative to the last available observation, so the 2–3 month GRACE-FO release latency and ERA5 availability at issue time are not modeled.

3.2.2 Metrics and tests.

The primary metric is pooled RMSE in standardized units, converted to a skill score $1 - \mathrm{MSE}_A/\mathrm{MSE}_B$ against a stated reference. Pooled model comparisons use the Diebold–Mariano test (Diebold and Mariano, 1995) on cross-basin monthly mean loss differentials, with the small-sample correction of Harvey et al. (1997) and Newey–West HAC variance at maximum lag $\max(h-1, 1)$ (Newey and West, 1987). Confidence intervals use a moving-block bootstrap (Künsch, 1989) with block length $\max(3, h)$ and 2000 draws; the block covers the lag-$(h-1)$ dependence of overlapping h-step forecasts. Per-basin significance uses per-basin DM statistics on the scored test months (83 at lead 1 declining to 78 at lead 6 as the shifted target runs off the record) with Benjamini–Hochberg false-discovery-rate control at $q = 0.10$ (Benjamini and Hochberg, 1995), following the field-significance reasoning of Wilks (2016); we always report helped and hurt basins separately.

3.3 Baseline ladder and the observation-noise-aware benchmark

The baseline ladder comprises: zero-anomaly climatology; persistence (last observed anomaly); damped persistence in two variants (an AR(1) damping coefficient fit per basin, and a regression-estimated variant), of which we always score against the stronger variant at each lead; a pooled ridge on own-basin lagged anomalies ($\alpha = 1$, lags 0, 1, 2, 5, 11 months); the same ridge with climate-index lags; and a per-basin ridge.

The benchmark this paper argues for models each basin's deseasonalized residual y_t as an AR(1) latent signal x_t observed with additive white noise:

$$x_t = \rho x_{t-1} + \epsilon_t, \qquad \epsilon_t \sim \mathcal{N}(0, q), \tag{1}$$

$$y_t = x_t + \eta_t, \qquad \eta_t \sim \mathcal{N}(0, r), \tag{2}$$

with (ρ, q, r) estimated per basin by maximum likelihood (L-BFGS-B, multiple starts) on pre-test data. The h-month forecast is $\hat{y}_{t+h|t} = \rho^h x_{t|t}$, damped persistence of the *filtered state*. The point forecast depends on ρ and the noise ratio q/r only; the filter handles the mission gap and missing months by skipping update steps, which inflates state uncertainty without biasing the state. We call this forecast the *Kalman forecast*. Two further members of the same "filtering class" reuse the filtered state: the *ridge on filtered states*, a pooled ridge on lagged filtered states, and the neighbor-augmented variants of Sect. 3.4. We emphasize what this benchmark is not: it assimilates nothing into any model and is unrelated to the GRACE data-assimilation literature reviewed by Springer et al. (2026); it is a forecast reference with three scalars per basin.

3.3.1 Identification caveat, and the $r = 0$ ablation.

The decomposition into signal variance q and observation-noise variance r is a modeling choice, not a measurement: high-frequency hydrology, non-AR(1) dynamics, and decomposition error are absorbed into r alongside true instrument noise, and in a substantial minority of basin–fold fits the likelihood pushes r to its lower boundary: all 1170 basin–fold fits converge, but 259 of them land at $r < 10^{-6}$, and 31 basins sit at the boundary in all five folds. So r is not, per basin, a clean instrument-noise estimate. Whether the benchmark's *margin* is nonetheless the filtering term is a separate, answerable question, and an ablation answers it. Refitting the identical model class by the identical MLE with r fixed at zero forces the update gain to one, so the "filter" tracks the raw observation and the model degenerates to damped persistence with MLE ρ. This removes the entire margin and more. The free-r filter beats its $r = 0$ twin by $+5.4\%$ at lead 1 rising to $+12.7\%$ at lead 5 (DM $p < 10^{-13}$ at every lead), while the $r = 0$ twin *trails* the tuned damped-persistence baseline at leads 2–6 (-0.9 to -11.6%): the AR(1) structure alone, without the noise term, is worth nothing beyond persistence at any lead past the first. What is filtered out doubtless mixes instrument noise with fast unmodelled hydrology; the ablation attributes the margin to the act of filtering the observation before extrapolating, not to a specific noise source.

3.4 Neighbor graphs, controls, and the jump screen

3.4.1 Graphs.

For each target basin and fold we select one source basin (top-1) by: training-window Pearson correlation of the TWSA series (*correlation graph*, the headline treatment); correlation of the source's one-month-lagged series with the target (*predictive-lag graph*); or great-circle proximity (*geographic graph*). Neighbor information enters as the lagged, forward-propagated filtered state of the source basin appended to the target's feature set. Distance-restricted variants exclude sources within 300, 500, or 1000 km.

3.4.2 Degree-matched random graphs.

This is the primary control throughout the paper. Every neighbor model is compared against 20 or 50 placebo graphs that preserve in-degree but draw source basins uniformly at random. The placebo models are refit end-to-end, so the comparison is capacity-matched; beating all N draws gives an empirical rank $p_{\mathrm{rank}} = 1/(N{+}1)$. Two properties of this statistic must be kept in view. First, p_{rank} is bounded below by $1/(N{+}1)$, so at $N = 20$ every "significant" cell is pinned at 0.0476 and the rank p understates the evidence; where the margin matters we therefore also report the real graph's position in placebo standard deviations. Second, the placebo draws are redrawn independently for each fold and lead, but seeds within a cell share the cell's draws, so per-seed rank results at a given lead are re-scorings of the same 20 graphs (Sect. 5.5).

3.4.3 IAAFT surrogates.

Iterated amplitude-adjusted Fourier transform surrogates (Schreiber and Schmitz, 1996, 2000) preserve each basin's marginal distribution and power spectrum while destroying cross-basin phase alignment. The transform runs on the full monthly grid with GRACE gap months linearly interpolated and the gap pattern re-imposed afterward, so the preserved spectrum belongs to the true monthly time axis rather than a gap-compressed one. The *real* selected graph is reused for all 99 surrogate datasets: the tested null is that the selected neighbor's time alignment carries no information, with selection itself covered by the random-graph placebos. Surrogates are generated from the full record, test months included: no information flows into the real model, and the null draws share the real series' full-record statistics, which keeps the null closely matched to the real model. Because the surrogate distributions are estimated with test-period observations, however, this is a transductive sensitivity rather than a fold-pure prospective control, and we do not claim the design is conservative.

3.4.4 Conditioning.

Climate-index conditioning gives ENSO and IOD lags to every model in a comparison; ERA5 conditioning gives all 11 local-meteorology variables (lags 0–2, deseasonalized, train-standardized, winsorized at $\pm 10\sigma$) to every model, while placebos re-randomize only the neighbor. The test is one-directional: if conditioning leaves the neighbor estimate unmoved, the neighbor is not acting as a proxy for shared local weather. It does not, on its own, establish that the neighbor carries a signal. The winsorization guard exists because near-zero training variance in arid basins otherwise detonates standardized ERA5 features (the 2023 Libya flood reaches several orders of magnitude beyond the training standard deviation).

3.4.5 Footprint-sharing stratification.

The physical leakage quantity a neighbor effect could be an artifact of is not basin size but the degree to which a basin's mascon footprint reaches beyond the basin. We therefore define, per basin, a *footprint-sharing score* (named `contamination` in the archived files): the share of the basin's mascon-weighted signal contributed by land outside the basin, $\sum_t (w_t/W)\,(1-f_t)$, where w_t is the basin's weight on native tile t and f_t is that tile's fill fraction from within the basin. Footprint sharing and area are nearly independent axes (Spearman -0.325), so stratifying on one is not a restatement of the other. We report footprint-sharing terciles, a $90{,}000\,\mathrm{km}^2$ area split, and the 2×2 cross of the two. One property of the metric bounds its interpretation: it treats the union of all 284 mask units as "land", so the ocean half of a coastal tile is not counted as foreign. Island and coastal basins therefore score low, and the metric is a *lower bound* on footprint sharing.

3.4.6 Jump screen.

A 6σ outlier screen flags basins with step-like jumps and removes them before the neighbor comparison, in a train-only version and in a test-informed variant kept as a post-hoc diagnostic only. On the corrected pipeline the train-only screen flags 2 basins (Taiwan; Northern Kordofan) and leaves the lead-1 neighbor comparison at $+0.21\,\%$ against its own-ridge reference, still ahead of all 50 placebo draws ($p_{\mathrm{rank}} = 0.02$); the test-informed variant flags 27 and gives $+0.08\,\%$, likewise 50/50.

3.5 Learned models

All learned models predict the residual of the Kalman forecast (the final prediction is the Kalman forecast plus the learned correction), and each is paired with a ridge "twin" on identical features so that architecture and information content are never conflated. Unless stated otherwise, deep models are run with multiple seeds and reported as seed ensembles.

3.5.1 Nonlinear heads.

Gradient-boosted trees and multilayer perceptrons on the same feature sets as the ridges.

3.5.2 Residual MLP.

Stage one: the ridge twin on own-basin features. Stage two: a small MLP (hidden layers 64–32, early stopping) trained on the stage-1 *training* residual using only the neighbor's propagated filtered-state lags. Three seeds.

3.5.3 LSTM.

A shared-encoder LSTM (Hochreiter and Schmidhuber, 1997) over 12-month windows of the own filtered state concatenated with the 11 ERA5 channels, with a time-ordered $\sim 85/15$ train/validation split for early stopping; two seeds. Variants add the neighbor state as a thirteenth input channel.

3.5.4 The stacked system: LSTM plus a residual-correction stage.

Stage one: the own-state + ERA5 LSTM above, no neighbor. Stage two: an MLP trained on the stage-1 training residual from the propagated neighbor state only. Placebo graphs re-randomize only the stage-2 neighbor and ride the stage-1 network of the matching seed, so the placebo comparison is like-for-like within each seed. This construction matters: scoring one seed's model against another seed's placebos introduces a handicap several times larger than the placebo spread itself. The same training run also produces a neighbor-free *own-state control*, which sends each basin's *own* propagated state through the identical stage-2 path – same stage-1 network object, same stage-2 model, capacity, and seed – so that only the information source differs. We report it as a diagnostic rather than as the primary control, for reasons given in Sect. 4.5.

3.5.5 Graph attention network.

A one-layer GAT (Veličković et al., 2018) with one- and two-hop variants, three seeds, included to test learned message passing.

Table 1 collects the model names used in the rest of the paper; each name is used for exactly one model throughout.

3.6 Comparison protocol for GRACE-FCast

The Li (2025) hindcast is aggregated to our basin geometry by exact $0.25° \rightarrow 1°$ cell-containment weights; 7 island basins with zero coverage in their land mask drop out, leaving **227 matched basins**. Their forecasts are moved into each fold's deseasonalized standardized space using our train-window climatology plus a per-(fold, basin, lead) mean offset estimated on pre-test data only. The matched sample is the identical set of (basin, target-month) rows for every model at every lead: 60 test months, folds 1–4; fold 5 lies entirely beyond their final initialization. This gives a complete rectangle of $227 \times 60 = 13{,}620$ rows per model at each lead.

Two protocol asymmetries favor their system and are carried as caveats wherever it wins: their training uses the Yin et al. (2023) gap reconstruction (hindsight information across the 2017–2018 gap), and their product re-extrapolates seasonal and trend components monthly whereas our climatology is fold-frozen – a difference worth about 0.07–0.09 in skill units at leads 4–6, bounded by their non-seasonal variant, which still wins those leads.

Table 1. Model names used throughout the paper, with the defining subsection. Appendix A maps each name to its identifier in the archived result files.

Name	Definition
Climatology	Zero-anomaly forecast (Sect. 3.3).
Persistence	Last observed anomaly (Sect. 3.3).
Damped persistence	Last anomaly damped toward zero; two per-basin damping variants, always scored against the stronger at each lead (Sect. 3.3).
Pooled ridge	Ridge on own-basin lagged anomalies, one model for all basins; a variant adds climate-index lags (Sect. 3.3).
Per-basin ridge	The pooled ridge's features, refit separately for each basin (Sect. 3.3).
Kalman forecast	The proposed benchmark: damped persistence of the filtered state from the per-basin AR(1)-plus-observation-noise model (Sect. 3.3).
Ridge on filtered states	Pooled ridge on lagged filtered states (Sect. 3.3).
Neighbor ridge	Ridge on filtered states plus one selected neighbor's propagated filtered state; graph-construction, distance-restricted, and conditioned variants (Sect. 3.4).
Own-basin ridge / ERA5 ridge	Ridge twins on the basin's own features, without and with the ERA5-Land lag features (Sect. 3.5).
GBM head / MLP head	Gradient-boosted trees and multilayer perceptron on the same features as the corresponding ridge twin (Sect. 3.5).
Residual MLP	Two-stage model: a ridge first stage plus a small MLP trained on the stage-1 training residual; variants differ in the features each stage receives (Sect. 3.5).
LSTM	Shared-encoder LSTM over 12-month windows of the filtered state and the ERA5-Land channels; the neighbor-channel variant adds the neighbor state as a thirteenth input channel (Sect. 3.5).
Stacked system	The LSTM as stage 1 plus a neighbor-only MLP correction trained on the stage-1 training residual (Sect. 3.5).
Own-state control	The stacked system with stage 2 fed the basin's own propagated state instead of a neighbor's; a diagnostic, never a skill number (Sects. 3.5 and 4.5).
GAT	One-layer graph attention network, one- and two-hop variants (Sect. 3.5).
GRACE-FCast	The published forecast system of Li and Kusche (2026); full-signal and non-seasonal products (Sects. 2.4 and 3.6).

Table 2. Pooled skill (%) versus the stronger damped-persistence variant at each lead, matched rows across all models (234 basins, five folds; $n = 19{,}422/19{,}188/18{,}954/18{,}720/18{,}486/18{,}252$ at leads 1–6, constant across models; see Sect. 3.2). Positive is better. The reference is ρ-damping at lead 1 and regression damping at leads 2–6, whichever is stronger; the ρ-damped variant alone degrades to -6.4 to $-12.0\,\%$ at leads 3–6 and scoring against it would flatter every other row. The Kalman forecast is damped persistence applied to the filtered state rather than to the raw observation; "ridge on filtered states" is a pooled ridge on lagged filtered states. DM p values for the two filtering-class rows are $\leq 8 \times 10^{-7}$ at every lead.

Model	h=1	h=2	h=3	h=4	h=5	h=6
Climatology (zero anomaly)	-125.6	-70.7	-58.9	-51.4	-42.7	-33.6
Persistence	-20.7	-22.7	-21.4	-19.5	-16.0	-15.8
Damped persistence (stronger variant)	0.0	0.0	0.0	0.0	0.0	0.0
Pooled ridge (own lags)	$+1.2$	$+4.8$	$+5.1$	$+3.6$	$+3.2$	$+2.5$
Pooled ridge + climate indices	$+1.0$	$+4.5$	$+5.1$	$+3.7$	$+3.2$	$+2.4$
Per-basin ridge (own lags)	$+3.1$	$+5.9$	$+4.3$	$+2.1$	$+0.4$	-0.3
Kalman (filtered-state persistence)	$\mathbf{+5.0}$	$\mathbf{+8.8}$	$+5.6$	$+3.1$	$+2.5$	$+3.6$
Ridge on filtered states	$+4.8$	$+7.9$	$\mathbf{+6.0}$	$\mathbf{+5.4}$	$\mathbf{+5.1}$	$\mathbf{+5.2}$

Skills are recomputed on matched rows from the archived per-prediction files; the archived source file for every table and figure is listed in Appendix A.

4 Results

4.1 The benchmark: persistence of the filtered state

Table 2 and Fig. 1 show the matched-row baseline ladder. The filtering class supplies the strongest own-basin reference at every lead. The bare Kalman forecast beats the stronger damped-persistence variant by $+4.98\,\%$, $+8.79\,\%$, $+5.62\,\%$, $+3.07\,\%$, $+2.55\,\%$, and $+3.63\,\%$ at leads 1–6 (DM $p \leq 8 \times 10^{-7}$ everywhere). It beats the per-basin ridge at five of six leads ($+1.4$ to $+3.9\,\%$, DM $p \leq 0.013$); at lead 4 the margin is $+1.04\,\%$ and the DM test does not reject ($p = 0.078$), so we do not count that lead. (The moving-block bootstrap interval at lead 4 excludes zero; the DM test is the pre-specified criterion and the one we report.) Against the *pooled* ridge the filter is significant at leads 1–2 only ($+3.86\,\%$ and $+4.21\,\%$); at leads 3–6 the two are not separable, and at leads 4–5 the filter is nominally behind ($-0.59\,\%$, $-0.64\,\%$, both n.s.). At leads 3–6 the filtering class is instead carried by a pooled ridge on lagged *filtered* states, which beats the per-basin ridge by $+3.4$ to $+5.4\,\%$ at leads 4–6 ($p \leq 2.1 \times 10^{-6}$): the signal/noise separation, not the ridge head, is what carries the class.

Three readings of Table 2 matter for the field. First, the gap between raw persistence and damped persistence (16–23 %) is the familiar cost of not damping. The further 5.0–8.8 % from damping the filtered state instead of the observation is the filtering-class margin, and the $r = 0$ ablation of Sect. 3.3 identifies it as the noise-filtering term specifically: the same model class with the noise variance removed gives the margin back and more. The margin is of the same order as many published model-over-persistence margins. without noise separation sit *between* the two persistence variants and the filter: the per-basin

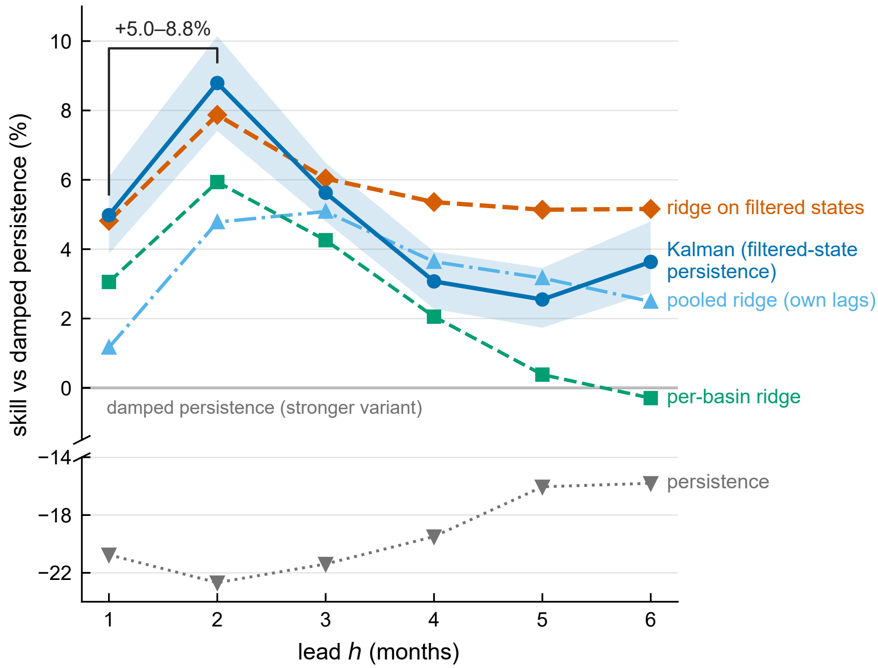

Figure 1. The benchmark ladder as skill versus lead. Pooled skill (%) relative to the stronger damped-persistence variant at each lead for the models of Table 2, with a block-bootstrap 95 % confidence band on the Kalman curve. The gap between damped persistence of the raw observation (zero line) and damped persistence of the Kalman-filtered state is the gain from filtering the anchor observation: 5.0–8.8 % at leads 1–2. Conventional statistical models (pooled and per-basin ridge) sit between the two persistence classes at every lead. Raw, undamped persistence is far weaker and is shown in the broken lower panel.

ridge – five lag coefficients per basin – never reaches the three-parameter filter at any lead, and by lead 6 it has fallen below damped persistence altogether. Third, the benchmark is nearly free: it adds no data requirements and two lines of state-space code, so we propose it as the default persistence-class reference for deseasonalized GRACE basin forecasting. The reading is constructive: the ladder does not say learned models are useless; it says the bar they must clear sits higher than the field's standard reference, and the rest of this section is a map of what clears it.

4.1.1 Protocol sensitivity.

The margin is insensitive to two evaluation-protocol choices that we tested directly – the calendar-month assignment of the two irregularly centered CSR solutions and the definition of fold membership on issue rather than target dates (Sect. 3.2): across the two conventions the filter's margin over damped persistence changes by at most 0.7 percentage points at any lead and its sign and ordering nowhere.

4.2 What beats the filter: forcing, and a longer window of forcing

4.2.1 Exogenous forcing is the largest single skill source, and it is a lead-1 effect.

Adding ERA5-Land lag features to the own-basin ridge yields $+4.62\,\%$ at lead 1 (95 % CI $+1.9$ to $+6.8$, DM $p = 1.0 \times 10^{-4}$) – an order of magnitude larger than anything the neighbor channel of Sect. 4.4 produces at the linear tier. Beyond lead 1 it is not detectable: $+1.39\,\%$ at lead 2 ($p = 0.092$) and $+0.33\,\%$ at lead 3 ($p = 0.61$). Forcing supplied to a linear head therefore buys one month; carrying it further requires the longer forcing window below.

Three decompositions locate the gain. By variable (Table 3), the information is spread across the collinear moisture variables rather than owned by any one of them: evaporation carries the only unique contribution that survives Bonferroni correction across the eleven leave-one-out tests, the top two soil-moisture layers are the strongest standalone predictors, and small negative unique terms for collinear variables appear as expected. By fold, the gain is robustly positive in four of five test windows ($+6.0$ to $+8.5\,\%$) and fails only in the window spanning the 2022–2023 triple-dip La Niña – the one climate state in the record where reanalysis forcing misinformed the model. By geography, the gain is a northern-hemisphere phenomenon: about $+6$ to $+7\,\%$ across North America, Europe, and Asia, and statistically zero in South America and Africa (Table 3). Per basin, 23 of 234 survive FDR control (21 helped, 2 hurt). Sect. 5.3 builds on the mirror-image relation between this geography and the neighbor signal's: ERA5 is weakest exactly where the neighbor signal is strongest.

4.2.2 A longer window of forcing extends the gain; sequence architecture does not.

On identical own-basin features, gradient boosting and MLP heads never beat their ridge twins; across every feature set the sole nominal exception is the gradient-boosted head on the neighbor-augmented features at lead 2 ($+1.12\,\%$, $p = 0.019$), which no other lead or head supports. With ERA5 aboard, the GBM is statistically tied at leads 1–2 ($+0.14\,\%$, $p = 0.77$; $+0.60\,\%$, $p = 0.13$) and significantly *worse* at lead 3 ($-0.71\,\%$, $p = 0.037$), while the MLP is tied at lead 1 ($-0.53\,\%$, $p = 0.30$) and significantly worse at leads 2–3 ($-2.56\,\%$ and $-3.14\,\%$, $p \leq 7.2 \times 10^{-5}$). This is the within-study analogue of Nie et al. (2025).

One model class appears at first to be an exception, and dissecting it attributes the gain to history length rather than architecture. An LSTM over 12-month windows of filtered state plus ERA5 channels beats the ridge by $+2.17\,\%$, $+2.36\,\%$, and $+1.43\,\%$ at leads 1–3 (two-seed ensemble, DM $p \leq 4.6 \times 10^{-5}$) – but that ridge sees only issue-time lag features, so the pair confounds the architecture with the amount of history it reads. Equalizing the information settles the attribution. A ridge over the identical 12-month history, flattened to a plain feature vector, gains $+3.34$, $+5.01$, and $+3.49\,\%$ over the lag-feature ridge – more than the LSTM's entire margin – and head-to-head the LSTM *loses* to it at every lead ($-1.20\,\%$, $p = 0.037$; $-2.80\,\%$, $p = 1.8 \times 10^{-10}$; $-2.13\,\%$, $p = 3.8 \times 10^{-9}$). Because the LSTM holds out its last 15 % of training months for early stopping, we also refit the flat ridge on the LSTM's exact training rows; the comparison is essentially unchanged, though the lead-1 margin is then not significant on its own ($p = 0.094$; leads 2–3 remain $p < 10^{-6}$). An MLP on the same flattened history loses to the flat ridge as well (-3.08 to $-2.26\,\%$). The apparent sequence-modeling gain is a history-length gain: twelve months

Table 3. Attribution of the lead-1 ERA5-Land gain ($+4.62\,\%$ pooled skill over the own-basin ridge). Left block: unique contribution of each variable (skill lost when it is dropped from the full set; Diebold–Mariano drop-test p) and standalone skill (that variable's lags alone; p against the no-ERA5 ridge). Right block: the same pooled gain split by continent. Fold-wise values appear in the text.

Variable	Unique (pp)	p	Solo (%)	p
Precipitation	$+0.57$	0.52	$+1.64$	0.22
Evaporation	$+1.45$	0.002	-1.19	0.042
Total runoff	$+0.31$	0.002	$+0.23$	0.37
Surface runoff	$+0.01$	0.53	$+0.97$	0.002
Subsurface runoff	-0.00	1.00	$+0.14$	0.56
Soil moisture, layer 1	-0.16	0.037	$+2.29$	$< 10^{-3}$
Soil moisture, layer 2	$+0.15$	0.038	$+2.22$	$< 10^{-3}$
Soil moisture, layer 3	$+0.17$	0.004	$+0.71$	0.008
Soil moisture, layer 4	$+0.42$	0.007	$+0.34$	0.023
2-m temperature	-0.08	0.002	$+0.25$	0.14
Snow water equivalent	$+0.06$	0.65	$+0.97$	$< 10^{-3}$
Continent	**n**	**Skill (%)**	**p**	
North America	46	$+7.05$	$< 10^{-3}$	
Europe	30	$+6.86$	$< 10^{-3}$	
Asia	79	$+5.99$	$< 10^{-4}$	
Australia–Oceania	14	$+3.39$	0.053	
South America	27	$+1.47$	0.66	
Africa	38	-0.39	0.87	

of forcing carries real information, the linear reader extracts the most of it, and the sequence architecture pays a premium for re-learning what the flattening gives away for free.

Without ERA5, the LSTM does not durably beat the ridge: $+0.32\,\%$ at lead 1 ($p = 0.10$), $+0.93\,\%$ at lead 2 ($p = 1.5 \times 10^{-4}$), and significantly *worse* at lead 3 ($-1.03\,\%$, $p = 0.008$) – the filter has already extracted most of the temporal structure of the TWSA series itself, and the usable history is the forcing's, not the TWSA's. A one-layer graph attention network never beats its ridge twin at any seed, lead, or graph variant (-0.90, -0.24, $-0.61\,\%$ at leads 1–3 against its own-graph twin), and its two-hop neighborhoods are never better than one-hop. An exhaustive scan of the GAT variants turns up exactly one nominal win ($+0.65\,\%$ at lead 2, $p = 0.035$, ensemble only), which no individual seed supports and whose placebo ranks are poor; we read it as noise. Learned message passing at n=234 basins with $\sim$65 training months per fold costs more than it captures.

Table 4. Head-to-head skill (%) of our stacked system (two-seed ensemble) versus the GRACE-FCast full product (Li and Kusche, 2026; Li, 2025) on the matched sample (227 basins, 60 months, 13,620 rows per lead). Positive means our system is better. Per-basin wins: basins with lower MSE for our system, of 227.

Lead	Skill (%)	95 % CI	DM p	Per-basin wins
1	$\mathbf{+20.0}$	$[+8.1, +30.5]$	3.4×10^{-4}	137
2	-3.5	$[-16.5, +7.6]$	0.50	82
3	-12.9	$[-24.0, -3.5]$	0.015	71
4	-17.4	$[-30.2, -8.3]$	0.0021	65
5	-23.6	$[-39.0, -15.3]$	1.3×10^{-4}	59
6	-30.3	$[-46.2, -22.8]$	1.5×10^{-6}	59

Archived source files are listed in Appendix A.

4.3 The crossing: two published systems on observed TWSA

Table 4 and Fig. 2 report the matched-protocol comparison between our best system (the stacked system, Sect. 4.5) and GRACE-FCast (Li and Kusche, 2026).

The pattern is an empirical crossing consistent with the classic initial-conditions-versus-forcing crossover (Wood and Lettenmaier, 2008; Shukla and Lettenmaier, 2011; Yossef et al., 2013), here measured for the first time on satellite-observed TWSA between two real, published, data-driven systems: our system wins lead 1 decisively, lead 2 separates neither system, and their margin widens monotonically from lead 3 (Table 4). We say "consistent with" rather than claiming a causal source decomposition: the two systems differ in architecture, training data, gap handling, seasonal treatment, and predictor sets simultaneously, so the lead profile, not any single design choice, is what is measured. The structural reading is nonetheless natural. Their LSTM takes no GRACE lags, so at lead 1, where the most recent observation is the single strongest predictor, the bare Kalman forecast alone – three parameters, no exogenous data – beats their full product by $+14.6\,\%$ ($p = 0.0039$). Against damped persistence on the matched sample the split is sharpest: our system scores $+11$ to $+17\,\%$ at every lead, whereas theirs starts nominally below damped persistence at lead 1 ($-11.1\,\%$, $p = 0.057$) and rises to $+34\,\%$ by lead 6. The lead-1 deficit should not be over-read: it is a pooled-magnitude effect (their product still beats damped persistence in 115 of 227 basins), and damped persistence is not a baseline available to their intended user, since the 2–3 month mascon latency their product exists to bridge withholds the anchor observation. The crossing is robust to their known protocol advantages: against their non-seasonal variant, which removes the monthly re-extrapolated seasonal/trend edge, the crossing moves from between leads 2 and 3 to between leads 3 and 4 but does not disappear.

It is worth noting that Li and Kusche (2026) document the same target-definition sensitivity from the other side, and quantify it. Scoring their own product at lead 1, they report correlation above 0.8 over approximately $72\,\%$ of global land for the full signal but only $20\,\%$ once the seasonal cycle and linear trend are removed, and they attribute the difference to the extrapolated components directly: including them "improves correlation while increasing the overall error". Most of the correlation in a full-

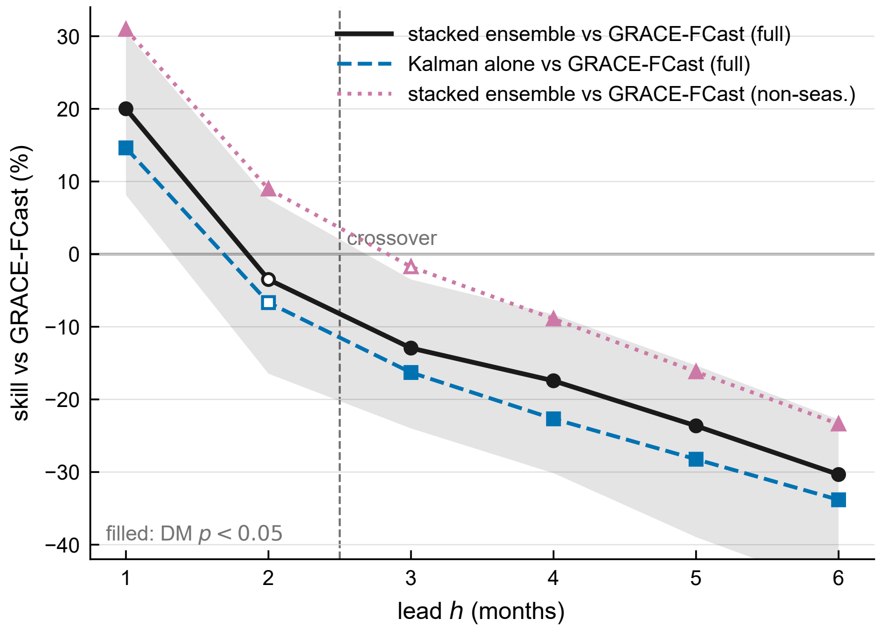

Figure 2. The initial-conditions/forcing crossover on observed TWSA. Skill (%) of our stacked system relative to the GRACE-FCast full product (Li and Kusche, 2026) as a function of lead on the matched sample (227 basins, 60 months), with block-bootstrap 95 % bands; a vertical line marks the crossover between leads 2 and 3. Secondary curves show the bare Kalman forecast against their product (structural lead-1 advantage of the filtered initial condition) and our system against their non-seasonal variant (the crossing survives removal of their seasonal/trend re-extrapolation edge). Filled markers denote Diebold–Mariano $p < 0.05$.

signal GRACE forecast score is therefore carried by terms that are extrapolated rather than predicted. That is our benchmark argument restated in their numbers, and it is why the deseasonalized target is the one on which the two systems can be compared at all.

Because no other published study reports our pooled deseasonalized metric, we also state physical units on the matched sample. Pooled RMSE favors our system at leads 1–3 (5.0–5.8 cm against their nearly lead-flat 7.6–7.7 cm), but that comparison flatters us: pooled centimeter error is dominated by a few high-amplitude Arctic glaciated basins where their product is weakest. The per-basin median, which those basins cannot dominate, hands over one lead earlier – 2.4 versus 2.7 cm in our favor at lead 1, then in their favor from lead 2. The earlier handover sharpens the complementarity reading: the filtering advantage is concentrated at lead 1 and in exactly the high-amplitude basins the pooled number overweights. The equal-weight standardized comparison of Table 4 remains our primary metric.

4.3.1 A descriptive best-of-both splice.

Splicing our forecasts at short leads onto theirs at long leads is a way to quantify the complementarity, and we report it as a descriptive sketch rather than a validated forecast product: the splice points would be chosen on the same test sample, several variants were examined without multiplicity control, and pooled DM differentials over all leads are diluted by the shared long-lead segment. The splice's gain lives at the one lead where it differs from their product: handing over after lead 1 improves on their forecast there by $+14.5\,\%$ using our filtered state and $+17.3\,\%$ using the ERA5-conditioned variant ($p \leq 0.004$), while row-pooled over all six leads the splice clears their product by well under a percentage point, because four of the six pooled leads *are* their product. The per-lead tests in Table 4 are the evidence either way; the splice only illustrates what a filtering-plus-forcing hybrid could deliver.

4.4 Cross-basin information at the linear tier: a null result with a footprint-sharing pocket

At the linear tier the cross-basin channel does not pay. Adding one correlation-selected neighbor's lagged filtered state to the ridge on filtered states – the pre-specified, capacity-matched paired comparison, in which both models share the same head and own-basin features – changes pooled lead-1 skill by $+0.31\,\%$ (95 % CI -0.09 to $+0.82$, DM $p = 0.199$) – not significant. At leads 2–6 the estimate is negative at every lead (-0.10, -0.19, -0.36, -0.35, $-0.32\,\%$), also without significance; the nearest thing to a significant result anywhere in this family runs *against* the neighbor (lead 5, $p = 0.070$). The estimate is also protocol-sensitive – under the target-date fold convention rejected in Sect. 3.2 the same contrast reads $+0.49\,\%$ with $p = 0.022$ – which is itself a caution for spatial-TWSA claims evaluated under looser protocols. What makes the null informative rather than merely inconclusive is the control battery the estimate was put through (Table 5 and Fig. 3), which no prior spatial-TWSA claim has faced, and which localizes the little that is there.

4.4.1 Two questions that must be kept apart.

The placebo evidence and the paired-contrast evidence answer different questions, and blurring them is how a null becomes an apparent finding. *Does correlation-selection beat a random neighbor?* At lead 1, yes: the real graph beats all 50 degree-matched random graphs ($p_{\mathrm{rank}} = 0.0196$), and so does the $\geq$300 km-restricted variant. The selection step is doing something real. *Does adding a neighbor beat not adding one?* No ($p = 0.199$). The first question is about graph construction; only the second is about skill, and only the second is the paper's claim. Beyond lead 1 both answers turn negative together: at leads 2–6 the real graph is beaten by 0 of 50 placebos, meaning the correlation-selected neighbor is worse than a randomly chosen basin. (The $\geq$300 km variant reaches 2 of 50 at lead 2; 0 of 50 elsewhere.)

4.4.2 Stratification (primary sensitivity): the effect is a footprint-sharing pocket, not a size effect.

Splitting the 234 basins at $90{,}000\,\mathrm{km}^2$ puts the whole lead-1 estimate in the small stratum (35 basins, $+1.033\,\%$, $p = 0.00065$) and none of it in the large one (199 basins, $+0.099\,\%$, $p = 0.715$), and that split is not a knife edge: every cut from 60,000 to $200{,}000\,\mathrm{km}^2$ gives the small stratum $+0.9$ to $+1.3\,\%$ and leaves the complement non-significant.

Reading this as GRACE's resolution limit, however, would mistake a proxy for the mechanism. The physical quantity is *footprint sharing* – how much of a basin's mascon-weighted signal comes from land outside the basin (Sect. 3.4) – and it is nearly independent of area (Spearman -0.325). Crossing the two axes separates them:

Area stratum	Footprint sharing	Basins	Lead-1 skill (%)	DM p
Resolved ($\geq$90,000 km^2)	high (top tercile)	57	$+1.161$	0.0045
Resolved ($\geq$90,000 km^2)	low/mid	142	-0.500	0.082
Small ($<$90,000 km^2)	high (top tercile)	21	$+1.166$	0.0071
Small ($<$90,000 km^2)	low/mid	14	$+0.913$	0.036

Footprint sharing, not basin size, is the stratifier that tracks the linear neighbor effect. Among basins GRACE comfortably resolves, the high-sharing stratum carries $+1.16\,\%$ – the same magnitude as in the small basins – while the low-sharing majority runs *negative*. One cell resists the pattern: the 14 small, nominally low-sharing basins also come in positive ($+0.913\,\%$, $p = 0.036$). This is the smallest cell in the table and the one the footprint-sharing score measures worst: the score counts only foreign *land*, so it understates sharing for the coastal and island basins that make up most of this cell (Sect. 3.4). Repeating the cross at CSR's published $200{,}000\,\mathrm{km}^2$ caution threshold (Sect. 2) sharpens rather than weakens the picture: basins above the cut with high sharing keep the effect ($+1.10\,\%$, $p = 0.033$), the low/mid-sharing majority above the cut is significantly *negative* ($-0.80\,\%$, $p = 0.014$), and the resisting small-and-clean cell shrinks to marginal ($+0.69\,\%$, $p = 0.057$). Contamination terciles reach the same conclusion without any threshold choice (low $-0.27\,\%$, mid $-0.18\,\%$, high $+1.16\,\%$, $p = 0.00051$). The pocket is short-lead as well as sharing-confined: for the ERA5-conditioned neighbor ridge the high-sharing tercile gives $+1.11\,\%$ at lead 1 ($p = 2.3 \times 10^{-4}$) and $+0.59\,\%$ at lead 2 ($p = 0.037$), then nothing at leads 3–6, while the mid and low terciles are negative at every lead.

Short-lead, confined to shared footprints, and absent where footprints are clean is the signature of leakage, not of hydrology. A centroid-distance exclusion does not remove shared mascons for such basins, so we rest no novelty claim on the linear pocket. Accordingly we describe this section as a characterization with null-model controls, not as isolation and validation of a signal: the controls tell us what the pooled number is *not* (chance graph structure, shared weather), the paired test tells us it is not distinguishable from zero, and the stratification tells us that the little that is there sits exactly where an instrument artifact would put it. Whether the same is true of the nonlinear correction is a separate question with a different answer, and Sect. 4.5 tests it directly.

4.4.3 Conditioning does not explain the effect away.

Giving every model in the comparison the 11 ERA5 local-meteorology variables barely moves the lead-1 point estimate ($+0.322\,\%$ conditioned versus $+0.308\,\%$ unconditioned), so whatever the estimate reflects, it is not a proxy for shared local weather. Both estimates are non-significant, however ($p = 0.115$ and $p = 0.199$), so conditioning rules out the shared-weather reading without rescuing the effect itself.

Table 5. The lead-1 neighbor estimate against its null-model battery. Each row states the control, what it tests, and the outcome. All effects are pooled skill versus the own-basin reference on identical rows. The estimate being controlled is itself not significantly different from zero ($+0.31\,\%$, $p = 0.199$), so these rows bound what the estimate could be rather than validating a demonstrated effect.

Control	Question	Outcome
Degree-matched random graphs (50)	Chance / capacity?	**passed at lead 1 only**: beats 50/50 draws ($p_{\mathrm{rank}} =$ .02); at leads 2–6 beaten by 0/50
IAAFT surrogates (99)	Univariate spectra rather than cross-basin timing?	**passed at lead 1**: beats 99/99 surrogates ($p_{\mathrm{rank}} =$ .01); at leads 2–6 the real model trails the surrogate distribution, matching the null there
$\geq$300 km source exclusion	GRACE mascon leakage?	unchanged ($+0.32\,\%$, 50/50 placebos)
$\geq$500 / $\geq$1000 km source exclusion	How far does it reach?	decays then dies ($+0.16\,\%$ at 500 km; $-0.60\,\%$ at 1000 km, beaten by 0/50 placebos)
ENSO + IOD conditioning	Shared teleconnections?	point estimate unmoved ($+0.310\,\%$ vs. $+0.308\,\%$); still ahead of all 50 draws; the pooled contrast is non-significant either way
Full ERA5 conditioning (11 variables)	Shared local weather?	point estimate unmoved ($+0.322\,\%$ vs. $+0.308\,\%$); both non-significant
Train-only 6σ jump screen	A few outlier basins?	2 flagged; screened estimate $+0.21\,\%$, still 50/50 vs placebos
Area stratification ($\geq$/$<$ 90,000 km^2)	Small-basin leakage receivers?	**not passed**: the whole pooled effect sits in the 35 small basins ($+1.03\,\%$, $p =$ .0007) and none in the 199 large ones ($+0.10\,\%$, $p =$ .72)
Footprint-sharing 2×2 (area $\times$ sharing score)	Is it size, or is it shared mascon footprint?	**not passed, and it is sharing not size**: high-sharing basins carry $+1.16\,\%$ whether or not GRACE resolves them; low-sharing resolved basins run $-0.50\,\%$

All rows are from the corrected issue-date pipeline; archived source files are listed in Appendix A.

The controls constrain what the lead-1 estimate could be without ever lifting it clear of zero. It is not chance graph structure – the real graph beats all 50 degree-matched placebos – and it is not shared local weather, since full ERA5 conditioning leaves it unmoved. The centroid-distance exclusions bound its spatial scale ($\sim$300–1000 km). But the footprint-sharing stratification places what is left exactly where an instrumental explanation predicts it, and a centroid-distance rule between basin *centroids* does nothing about two basins drawing on the same mascon. The reading most consistent with the whole battery is that at the linear tier the neighbor is largely acting as a second view of partly the same measurement – which would explain why the estimate is confined to shared-footprint basins and to leads 1–2, and why it is negative beyond 1000 km. We claim no hydrological transfer, and at the linear tier we claim no effect.

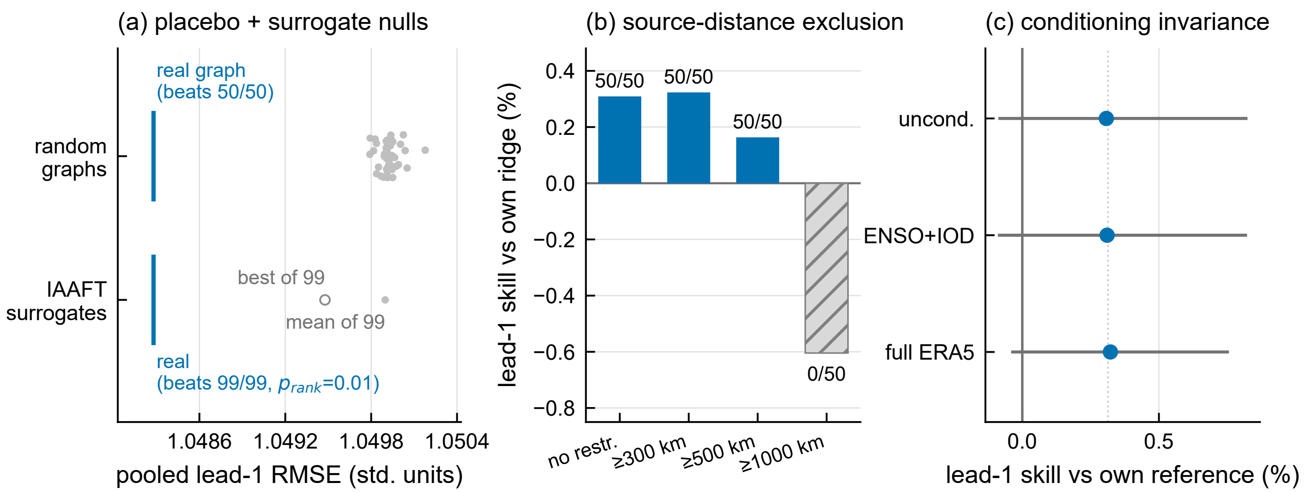

Figure 3. The lead-1 neighbor estimate against its null models. (a) Pooled lead-1 RMSE of the real correlation graph (vertical line) against its two null families: 50 degree-matched random graphs (dots) and 99 IAAFT surrogate pipelines (summary marks); the real graph beats every draw in both families. (b) Skill versus minimum allowed neighbor distance (no restriction, $\geq$300 km, $\geq$500 km, $\geq$1000 km), with placebo graphs beaten (of 50) above each bar: unchanged at 300 km (ruling out the crudest form of mascon leakage), halved at 500 km, and negative at 1000 km – bounding the spatial scale of whatever is being picked up. (c) The estimate unconditioned, with ENSO+IOD index conditioning, and with full 11-variable ERA5 conditioning: the point estimates are unmoved and every interval crosses zero, which is the point of the section.

4.4.4 Heterogeneity, and per-basin selection.

A pooled null does not mean a spatially uniform null: a stable pattern of helped and hurt basins can average to nothing. Two analyses probe this – a per-basin FDR map (Fig. 4) and a leave-one-fold-out selection experiment that applies the neighbor only where it helped in the other four folds. The latter is a measure of cross-fold stability in per-basin benefit, not an operational recipe: selection for early folds uses folds lying later in time, so a deployable version would have to select on past folds only. On the corrected pipeline the per-basin decomposition confirms the two-sided structure: 128 of 234 basins nominally favor the neighbor and 21 survive FDR control at $q = 0.10$ – **13 helped and 8 hurt**. The hurt set is not noise: it is the familiar great-river roster (Yenisey, Paraná, Yangtze, Amur) whose large, clean footprints invite spurious distant neighbors. The helped set skews African and coastal (Niger, São Francisco, Rukwa, Namibia, Zambezia, the West Nile Delta). Static covariates barely predict who benefits (random-forest cross-validated $R^2 = 0.05$; the top predictors are basin size and cross-fold neighbor stability), but past per-basin performance does: applying the neighbor only where it helped in the other four folds lifts the pooled lead-1 effect from $+0.31\,\%$ (n.s.) to $+1.12\,\%$ ($p < 10^{-9}$, $\sim$128 basins selected), 63 % of the in-sample oracle's $+1.78\%$, with the same pattern on the ERA5-conditioned variant ($+1.05\,\%$). Under the caveat above, this is a stability measurement, not a deployable recipe.

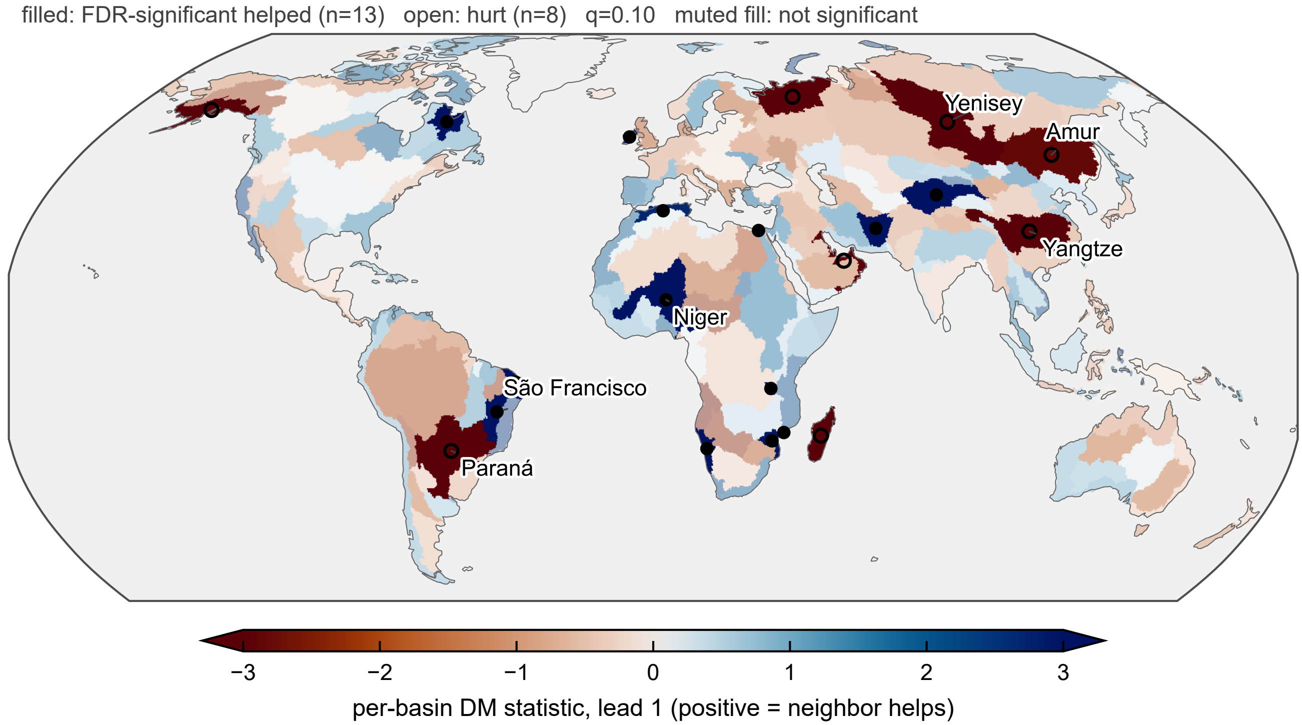

Figure 4. Geography of the lead-1 neighbor estimate. Global choropleth of the per-basin Diebold–Mariano statistic for the neighbor ridge versus its own-feature twin (234 basins; blue: neighbor helps, red: neighbor hurts), with basins passing Benjamini–Hochberg FDR control ($q = 0.10$; Wilks, 2016) shown at full saturation and marked by filled centroid dots; non-significant basins are muted. The map is two-sided by construction and should be read alongside the pooled null result of Sect. 4.4: a stable geography of helped and hurt basins is compatible with no pooled effect. Twenty-one of the 234 basins survive FDR control: the neighbor helps significantly in 13 and hurts significantly in 8. The extremes are as divided as the pooled null implies. Among the helped are the East Brazil South Atlantic coast ($\mathrm{DM} = -3.80$), the endorheic Tarim He–Lop Nur basin (-3.57), and the South Hudson Strait–Ungava Bay coast (-3.38); among the hurt are the Yenisey ($+4.26$), the South Gulf of Oman–West Arabian Sea coast ($+4.00$), and Madagascar ($+3.78$).

4.4.5 Architectures that should have enlarged the effect do not.

Structurally richer constructions all fail to enlarge the effect: predictive-lag neighbor selection loses to the own-basin reference at lead 1 ($-1.30\,\%$, $p = 0.0007$) and to all 50 placebo graphs – worse than a random basin; a single-latent fusion filter treating the neighbor as a second sensor of the same state loses to the plain Kalman forecast ($-2.14\,\%$); and a bivariate filter with correlated process noise is indistinguishable from it ($-0.02\,\%$, $p = 0.96$), ahead of 45 of 50 placebo draws but short of significance. Two-hop neighborhoods are no better than one-hop on the corrected pipeline (residual MLP $+0.13$/-0.28/$-0.15\,\%$, all n.s.; the GAT is significantly worse at lead 2), with one disclosed exception: the ERA5-conditioned residual MLP prefers two hops at lead 2 ($+0.82\,\%$ over one-hop, $p = 0.006$, all seeds positive), though chained against the own-basin twin the residual two-hop gain is only $\sim +0.2\,\%$. Together with the GAT null (Sect. 4.2), the linear-tier conclusion is that no standard linear or message-passing construction extracts usable cross-basin information from observed TWSA fields. That conclusion holds until the delivery mechanism changes, which is the subject of the next two subsections.

4.4.6 One nonlinear exception, and it is neighbor-specific.

The linear null is not the whole story even at the GRACE-only information set. A two-stage residual MLP fed *only* the neighbor's propagated filtered state beats its ridge twin on identical features by $+0.81$, $+2.15$, and $+2.17\,\%$ at leads 1–3 (three-seed ensemble, $p \leq 6.7 \times 10^{-5}$, every seed individually significant). The control that matters is the placebo family: every seed beats all 20 degree-matched random graphs at every lead (9 of 9 seed–lead cells: three seeds, three leads), and the placebo family itself pools to within 5×10^{-4} RMSE of the plain own-basin ridge. A junk neighbor through the same two-stage architecture adds nothing; the real neighbor adds up to two percent. This is the one place where cross-basin information survives on the same GRACE-only feature set that produced the linear null. The signal is nonlinear only, and it disappears – turning significantly negative – as soon as ERA5 forcing enters the model (-0.67, -0.62, $-0.55\,\%$, all $p < 0.05$). It supports the reading that cross-basin information matters most where forcing data are absent.

4.5 Representation decides: the neighbor is usable only as a propagated state

The delivery experiment holds the information *source* fixed – the same selected neighbor's Kalman-filtered state – and varies how it reaches the model. The input-channel variant receives the neighbor's 12-month filtered-state history as a full encoder channel, while the correction stage receives a single number per forecast, the neighbor's ρ^h-propagated current state. Delivery and representation are therefore confounded in that pair, so a third variant separates them: the correction stage fed the identical 12-month history. The answer is that representation, not delivery, is the operative variable. Feeding the neighbor's filtered-state history to the LSTM as one more encoder input channel yields no robust ensemble gain (Fig. 5): at lead 1 the two-seed ensemble gains $+0.36\,\%$ ($p = 0.090$), and at leads 2–6 it is negative throughout. The claim stops at "no robust ensemble gain": the per-seed picture is unstable enough that at lead 2 the two seeds disagree in sign, which is why the ensemble is the reported quantity. The representation-matched variant settles what that failure is about. Giving the correction stage the input variant's 12-month history in place of the propagated scalar – same stage, same capacity, same stage-1 network, only the representation changed – removes essentially the whole effect: the two-seed ensemble is not significant at any lead, its placebo ranks collapse, and against the propagated-scalar correction directly it loses at every lead (-0.87 to $-2.24\,\%$, all $p \leq 5.0 \times 10^{-3}$, both seeds). That is the same place the input-channel variant sits. Handed the neighbor's raw history, neither architecture extracts anything; handed the ρ^h-propagated state, the correction stage extracts $+0.9$ to $+2.0\,\%$. We therefore do not claim that the correction stage is what finds the signal. What the neighbor contributes is usable only once it has been propagated to the forecast target time, and the propagation is supplied by the filter of Sect. 4.1, not learned here: a deep encoder given 12 months of a neighbor's history does not recover, at this sample size, the alignment that one line of AR(1) propagation provides for free.

The stacked system trains the identical LSTM (own state + ERA5, no neighbor) and then fits a small MLP to the stage-1 *training* residual from the propagated neighbor state alone. Because challenger and reference share the same stage-1 network

Table 6. The stage-2 neighbor correction: skill (%) of the stacked system versus its own stage-1 LSTM (same network within seed; the contrast isolates the correction). The ensemble column is the reported quantity – predictions averaged across the two seeds per row, then scored – and the per-seed columns are the robustness check. Confidence intervals are moving-block bootstrap on the ensemble. The last column gives the number of 20 degree-matched random-graph correction placebos beaten, per seed.

Lead	Seed 0	Seed 1	Ensemble	95 % CI (ens)	DM p (ens)	Placebos beaten
1	$+0.64$	$+1.19$	$\mathbf{+0.91}$	$[+0.64, +1.22]$	1.1×10^{-10}	20/20, 20/20
2	$+1.33$	$+1.56$	$\mathbf{+1.45}$	$[+1.07, +1.91]$	2.3×10^{-10}	20/20, 20/20
3	$+1.29$	$+1.36$	$\mathbf{+1.33}$	$[+1.02, +1.69]$	1.9×10^{-9}	20/20, 20/20
4	$+1.26$	$+1.37$	$\mathbf{+1.32}$	$[+0.96, +1.76]$	3.1×10^{-8}	20/20, 20/20
5	$+1.51$	$+1.28$	$\mathbf{+1.42}$	$[+1.06, +1.84]$	6.2×10^{-9}	20/20, 20/20
6	$+1.90$	$+1.93$	$\mathbf{+1.96}$	$[+1.66, +2.36]$	1.2×10^{-13}	20/20, 20/20

Placebo graphs are redrawn independently for each fold and lead, so the twelve cells of the last column are separate draws; with 20 draws the rank statistic is nonetheless pinned at its $1/21$ floor (Sect. 5.5). Archived source files are listed in Appendix A.

within each seed, the contrast isolates the correction. Table 6 shows the result: the same information delivered this way adds $+0.91$ to $+1.96\,\%$ at *every* lead 1–6, in both seeds, with every confidence interval clear of zero.

Six properties give the correction its evidential weight.

First, and primarily, **the random-graph placebo.** This is the control the claim rests on. Its logic is that a placebo neighbor is a feature the stage-1 network has never seen – novel, like the real neighbor – but with the wrong identity, so it isolates identity from novelty. The correct statistic is the *increment* each stacked model adds over its own stage-1 network, compared with 20 random-neighbor corrections riding the matching stage-1 network. Placebo increments cost only -0.27 to $-0.05\,\%$ – a junk-neighbor correction buys nothing and pays only the small variance price of a second stage – while the real increment beats all 20 draws in all twelve lead–seed cells. We report the margin rather than the rank p, which is pinned at its $1/21$ floor: the real correction sits 6.8 to 18.6 placebo standard deviations beyond the placebo mean. Measured against the placebo median rather than against stage 1, the increment is $+0.92$ to $+1.96\,\%$ – two independently constructed references agreeing to within 0.1–0.3 percentage points at every lead. One accounting caveat: the draws are redrawn per fold and lead, but the two seeds within a lead are scored against the same draws, so the twelve lead–seed cells are six independent draw sets each scored twice, not twelve independent events.

Second, **the own-state control, a diagnostic rather than the primary control.** This neighbor-free variant feeds each basin's *own* propagated state through the identical feature path, same stage-1 network, same stage-2 model, capacity, and seed. Mechanically it is exact: the self-graph reproduces each basin's own propagated state to $\max|\Delta| = 0.000 \times 10^{0}$, and no zero-fill asymmetry exists between the two variants, since all 234 basins have a top-1 neighbor in every fold. The control actively *harms*: -1.16 to $-2.12\,\%$ across leads ($p \leq 2.5 \times 10^{-7}$). We use it as evidence that the stage-2 head has ample capacity to do damage, so the stacked system's gain is not a capacity artifact, and as evidence for the in-sample residual caveat below.

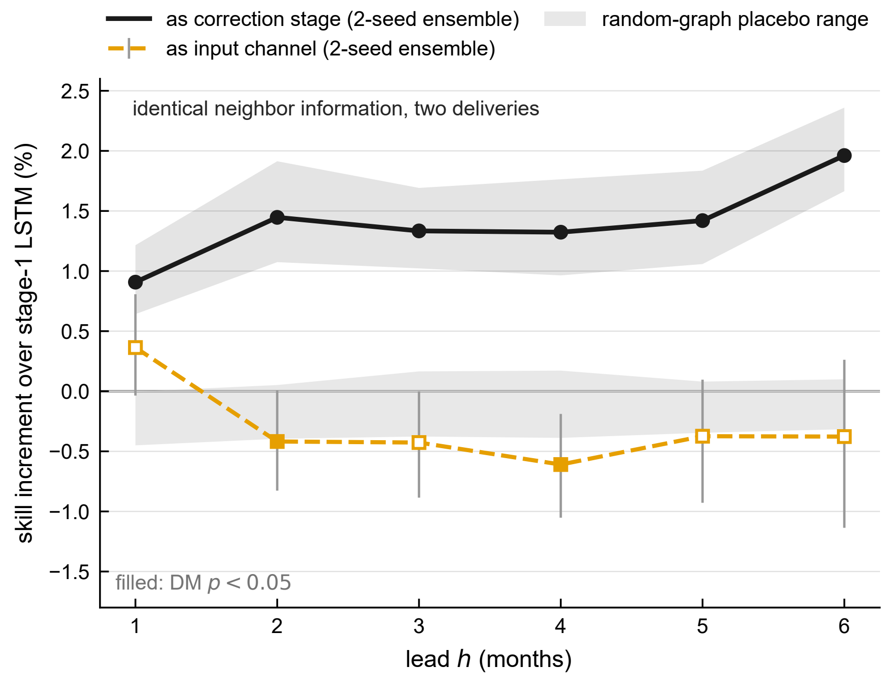

Figure 5. Representation decides whether the neighbor signal survives a deep model. Skill increment (%) of the neighbor information over the same own-state + ERA5 LSTM, by lead, for the two routes (both two-seed ensembles): as an additional encoder input channel ($+0.36\,\%$ at lead 1, negative at leads 2–6) and as a dedicated residual-correction stage ($+0.91$ to $+1.96\,\%$ at every lead). Filled markers denote Diebold–Mariano $p < 0.05$. Shaded band: the range of the 20 degree-matched random-graph placebo corrections per lead, each riding the matching stage-1 network and scored on both seeds; their mean cost of -0.27 to $-0.05\,\%$ is the pure variance price of adding a second stage. The real correction sits 6.8 to 18.6 placebo standard deviations beyond that band.

We do *not* use the neighbor-versus-own-state contrast as a skill number or as an "architecture term", because the control is too easy: its correction is anti-correlated with the stage-1 error (-0.08 to -0.16), and only 0.16–0.54 percentage points of its harm is variance cost, meaning 79–85 % of it is a systematically wrong-direction correction learned from in-sample residuals. A control that is actively harmful inflates any contrast drawn against it.

Third, **agreement between the two defensible references.** The correction is worth $+0.91$ to $+1.96\,\%$ against its own stage-1 network and $+0.92$ to $+1.96\,\%$ against the placebo median. Independent constructions landing on the same interval is the strongest single property of this result. Fourth, **the correction is positive in every fold as well as every stratum**: decomposing the ensemble increment over the five evaluation folds gives a positive value in all 30 fold–lead cells (minimum $+0.25\,\%$, in the earliest-freeze fold at lead 1; maximum $+2.63\,\%$), so no single evaluation period carries the result. At the basin level, 22 of 234 basins pass FDR control at lead 1 – 20 helped, 2 hurt, against 13 and 8 for the linear estimate – and the stack's per-basin

DM map rank-correlates with the linear neighbor map at $\rho = +0.49$ ($p = 1.4 \times 10^{-15}$): the two tiers largely agree on *where* the neighbor helps, and differ in how much of it they convert into skill.

Fifth, **lead stability that no other component has.** The stage-1 gain over the lag-feature ridge is gone by leads 5–6, where the LSTM is indistinguishable from that ridge ($-0.04\,\%$, $-0.11\,\%$). The correction does the opposite: it holds at $+1.42$ and $+1.96\,\%$ there, with the lead-6 value the largest in the column. At long leads it is the only architectural gain in the study that has not decayed.

Sixth, **it does not carry the leakage signature.** The obvious objection to any cross-basin claim on GRACE is that the "neighbor" is partly the same measurement, and Sect. 4.4 showed that this is exactly what the linear pocket looks like. We therefore put the stacked correction through the identical stratification – footprint-sharing terciles, the $90{,}000\,\mathrm{km}^2$ area split, and their 2×2 cross – at all six leads. The correction is positive in *all 60* strata-by-lead cells, and it does not concentrate in the strata a leakage artifact would favour:

Stratum	Basins	h=1	h=2	h=3	h=4	h=5	h=6
Sharing, low tercile	78	+0.69	+1.31	+1.29	+1.28	+1.31	+1.84
Sharing, mid tercile	78	+1.09	+1.58	+1.44	+1.33	+1.61	+2.01
Sharing, high tercile	78	+1.00	+1.48	+1.29	+1.35	+1.37	+2.03
Resolved × low/mid sharing	142	+0.89	+1.57	+1.46	+1.38	+1.49	+2.00
Small (<90,000 km^2)	35	+0.64	+0.88	+0.93	+0.82	+0.77	+1.05

All cells in the first four rows are significant ($p \leq 6.9 \times 10^{-3}$ in the low tercile, $p \leq 1.3 \times 10^{-4}$ elsewhere); the small-basin row is significant at every lead ($p \leq 0.044$) despite its n. The decisive cell is the fourth row: 142 basins that GRACE resolves and whose footprints are comparatively clean, where the *linear* effect is $-0.50\,\%$ and the correction is $+0.89$ to $+2.00\,\%$. The per-seed worst cell across every stratum and lead remains positive ($+0.22\,\%$, in a 21-basin cell).

"Positive everywhere" is not the same as "the same everywhere", and along the size axis it is not. Small basins gain *least*, and the gradient is real rather than visual: the size interaction is significant at every lead (lead-1 $p = 0.026$, lead-6 $p < 0.001$). We state that plainly because its direction is the reverse of the artifact's prediction – leakage would make the smallest, most footprint-shared basins gain *most*, and they gain least. Along the sharing axis the differences are not statistically distinguishable at any lead (high-vs-low interaction $p = 0.099$ at lead 1, with the point estimate trending toward *more* gain in high-sharing basins, and $p \geq 0.51$ at every other lead), which turns "not concentrated" from a description of the table into a tested statement.

The linear pocket and the stacked correction therefore have different signatures: one is confined to shared footprints and to leads 1–2, the other is positive in every stratum at every lead and, if anything, weakest where the artifact would be strongest. **Under this stratification the correction shows no detectable concentration where leakage would put it, and the size gradient runs opposite to the artifact's prediction.** That is the strongest statement the design supports: the footprint-sharing score is a lower-bound proxy (it counts only foreign land), the interaction test is a null result rather than proof of absence – with the lead-1 point estimate trending toward more gain under high sharing ($p = 0.099$) – and the stratification reuses the

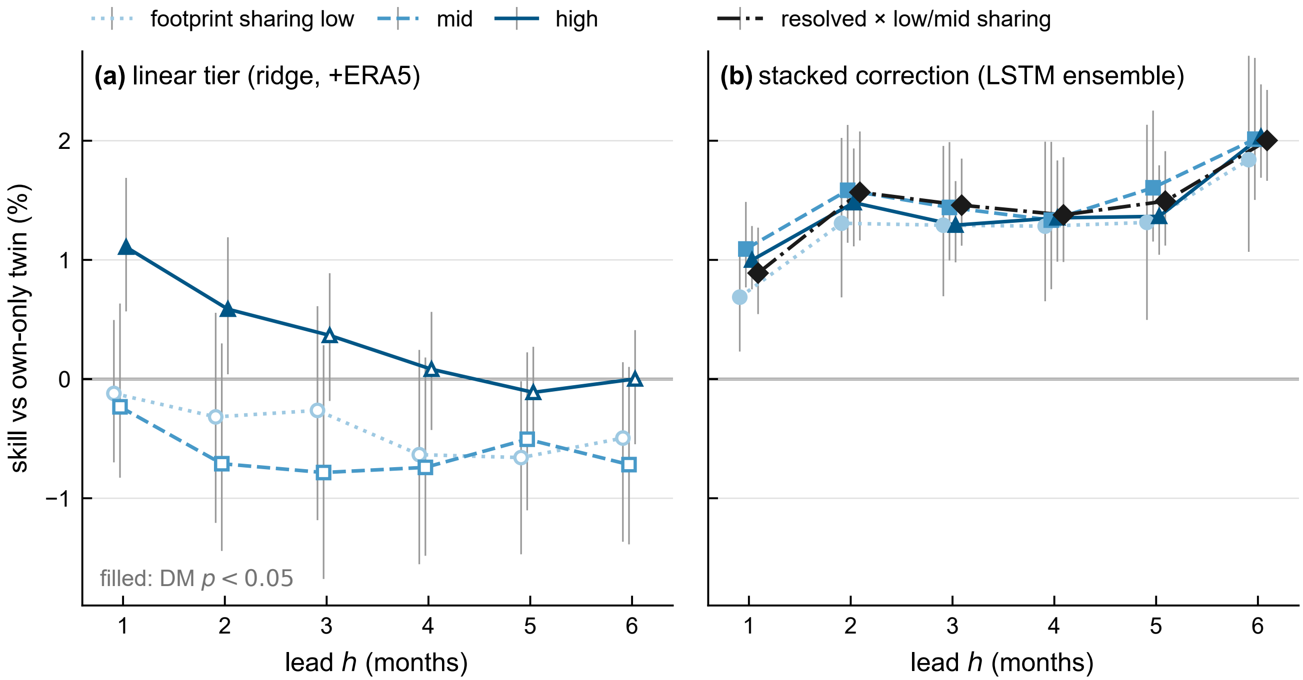

Figure 6. The linear neighbor effect carries a leakage signature; the stacked correction does not. (a) Linear tier: skill of the ERA5-conditioned neighbor ridge over its own-state twin by footprint-sharing tercile ($n = 78$ each), leads 1–6 – significant only in the high-sharing tercile and only at leads 1–2; everywhere else the point estimates are negative or indistinguishable from zero. (b) Stacked tier: the stage-2 neighbor correction over its own stage-1 LSTM (two-seed ensemble) for the same terciles plus the resolved, low/mid-sharing stratum ($n = 142$), with moving-block bootstrap 95 % intervals – positive in every stratum at every lead, including where the linear effect is negative. Both panels share the y-axis; the stratification reuses the headline's test months (Sect. 5.5).

headline's test months. Ruling on footprint sharing aside, the analysis does not identify what the neighbor is contributing, and we make no claim that it is a hydroclimatic teleconnection.

A diagnostic points the same way. The ERA5 residual MLP with neighbor features concatenated into its inputs – rather than given their own stage – is negative overall (-0.67, -0.62, $-0.55\,\%$), and its harm is concentrated in the *low*- and mid-sharing basins ($-0.89\,\%$, $p = 0.011$, in the resolved low/mid stratum) while being indistinguishable from zero where footprints overlap. Naive concatenation imports noise except where the neighbor is partly the same measurement; only the dedicated stage turns neighbor state into gains that are positive across every stratum.

Why does the same neighbor signal succeed in one form and not the other? Our first reading was gradient starvation (Pezeshki et al., 2021): in joint training a strong feature set (the forcing sequence) starves the gradient signal of a weak, partially correlated feature (the neighbor state), so the weak channel is ignored or fit to noise, and a dedicated residual stage rescues it by giving the weak feature its own objective. The representation-matched variant rules that out as the whole story. A dedicated stage fed the neighbor's raw 12-month history has its own objective and nothing else to fit, and it still recovers nothing (Sect. 4.5); if training dynamics were the binding constraint, that variant should have worked. Gradient starvation may still contribute to the input channel's instability, but it cannot explain a failure that survives the separation of objectives.

What distinguishes the variants that work is that the neighbor arrives already propagated to the forecast target time. $\rho_j^h x_j(t)$ is not a compression of the history; it is the neighbor's state carried forward by its own estimated dynamics, so the downstream model is handed a quantity that is temporally aligned with what it must predict. Recovering that alignment from 12 raw monthly values is a learning problem in its own right, and at this sample size – a few thousand rows per fold, one weak predictor among many – it is evidently not solved. The reading we can support is that the filter does more than clean the target: it also puts cross-basin information into the only form in which our models can use it. That places the propagation, not the architecture, at the center of the third contribution, and it is a claim about sample size and inductive bias rather than about optimization pathology. The residual-MLP experiments of Sect. 4.4 are consistent with it: they also feed propagated states, never raw history, where a neighbor-only residual MLP over the plain ridge beats its ridge twin on identical features by $+0.81$ to $+2.17\,\%$ while beating all 20 placebos in every seed–lead cell – and it does so despite the *linear* neighbor contrast on that same information set being a null. Two architectures, two information sets, one conclusion: the cross-basin signal is there, and it is reachable only when the neighbor is presented as a propagated state. A double-counting check completes the picture: stacking the correction on top of an encoder that *already ingests* the neighbor history yields $-0.46\,\%$ at lead 1 before drifting weakly positive at long leads ($+0.23$ to $+0.83\,\%$), never approaching the $+0.91$ to $+1.96\,\%$ the same correction earns over a neighbor-free encoder – consistent with the representation account: once the encoder has consumed the raw history, a second helping of the propagated state has little left to add, and the clean two-stage design remains the better construction.

4.5.1 Model selection and ranking caveats.

The stacked system is the nominal best model at leads 4–6 among all models evaluated there, and against the lag-feature linear system with the same predictor *types* it gains $+2.75$ to $+3.86\,\%$ at every lead. It is *not* the best model at short leads once history length is equalized: the flat 12-month ridge of Sect. 4.2 ties it at lead 1 ($-0.29\,\%$, $p = 0.58$) and beats it at leads 2–3 ($-1.31\,\%$, $p = 8.3 \times 10^{-4}$; $-0.77\,\%$, $p = 0.032$), and that model exists only at leads 1–3, so the long-lead ranking is untested against it. A leaderboard position is not the evidence, however, and one comparison in particular must be kept in view: the largest architectural number anywhere in this study is *neighbor-free* – a two-stage residual MLP on own-basin ERA5 features beats its ridge twin by $+3.35$, $+3.58$, and $+2.84\,\%$ at leads 1–3. Two-stage nonlinear correction is therefore worth more, in absolute terms, than the neighbor correction is, and a cross-architecture ranking cannot separate the two. On the corrected pipeline the direct contrast confirms it: the stacked system and the neighbor-free resMLP-ERA5 ensemble are a statistical tie at all three shared leads (-0.30, $+0.19$, $-0.10\,\%$; DM $p = 0.49$, 0.63, 0.81). The evidence that the neighbor adds skill is the controlled contrast of Table 6 – same stage-1 network, placebo-verified increment – not the cross-architecture leaderboard, and any deployment claim should be read accordingly. Moreover, the stacked architecture was chosen after inspecting earlier architectures' results on these same five folds, and no untouched evaluation period exists within the GRACE-FO record; its results are therefore **exploratory**, and the reported p values do not account for architecture-selection uncertainty. The stage-2 increment is protected by its seed-matched placebo battery, but confirmatory status awaits an untouched period or an independent product.

5 Discussion

5.1 What the benchmark result means, and does not mean

The benchmark result is an argument about reference classes, not a new forecasting method. Persistence-type references are supposed to represent "no information beyond memory"; for a noisy observable they leave the persistence class's own headroom unexploited, and that headroom on deseasonalized GRACE basin series is 5.0–8.8 % at the leads where most data-driven papers claim their skill. Any study that benchmarks against persistence, damped persistence, or climatology on observed TWSA – i.e., most of the literature cited in Sect. 1 – overstates the information its model has extracted by roughly this margin, whatever mixture of measurement error and unmodeled dynamics the filter is exploiting (Sect. 3.3). The fix costs three parameters per basin. This is the same benchmark-hygiene move that Niraula and Goessling (2021) made for the sea-ice edge via spatial damping, executed here via signal/noise separation, and it converges with three independent lines: persistence-class methods underrated at the coefficient level (Kankanige et al., 2026), statistical per-basin prediction competitive at basin level (Ahi and Cekim, 2021), and linear models unbeaten when autoregressive information is excluded (Nie et al., 2025). Our contribution to that conversation is an ablation-grounded attribution: with autoregressive information included, what separates the models the field builds is mostly how well they filter it – deleting only the noise term from the benchmark deletes the margin (Sect. 3.3). The ablation fixes the attribution within the model class; what the filtered component physically is – instrument noise, fast hydrology, decomposition error – remains unidentified (Sect. 3.3).

Two scope notes. The argument applies to *observed* TWSA; targets that are themselves model output, reconstructions, or gap fills (e.g., the HydroGlobe target of Nie et al., 2025, or the products of Humphrey and Gudmundsson, 2019; Yin et al., 2023; Mo et al., 2022) contain no satellite observation noise to filter, so neither the benchmark inflation nor its fix transfers. And our filter is not data assimilation (Springer et al., 2026): nothing is assimilated into any model, and no state but the basin's own scalar signal is estimated. The closest upstream machinery we know of is the integrated postprocessing of Zhang et al. (2025), which models the non-seasonal GRACE signal as a stochastic process and iteratively separates it from noise in the monthly solutions. That work shows signal/noise separation improves *extraction* of TWSA from the existing record; ours differs in pointing the separated state forward – using it to forecast, and to benchmark forecasts – so the benchmark claim here is not that filtering GRACE is new, but that forecast comparisons on observed TWSA are uninterpretable without it.

5.2 The crossover as a design map

The initial-conditions-versus-forcing partition is old (Wood and Lettenmaier, 2008; Shukla and Lettenmaier, 2011; Koster et al., 2010; Yossef et al., 2013; Pechlivanidis et al., 2020); what is new is its location on satellite-observed TWSA, measured between two published data-driven systems with paired tests. The crossover at $\sim$2 months is earlier than for streamflow in snow-dominated regions but consistent with TWSA's strong memory being concentrated in exactly the component the filter extracts. The practical reading is a design map, not a contest. Below the crossover, invest in filtering the initial condition: three parameters suffice, and further architecture buys little. Above it, invest in exogenous forcing and in a longer window of the forcing history. Our own evidence (Sect. 4.2) orders the forcing side: a ridge on issue-time forcing lags cannot carry ERA5 skill past lead 1, models reading a 12-month forcing window carry it to lead 4, and Li and Kusche (2026)'s system, which commits fully to this side, carries forcing-driven skill to lead 6 and beyond. (Whether the lead-4 step needs the sequence architecture rather than the longer window is untested; the flattened-history ridge exists only at leads 1–3.) That the linear lag-feature ERA5 gain stops at lead 1 sharpens the map: the useful part of forcing information at longer leads is not in the issue-time lags but in the depth of history a model reads. At basin level the two edges are uncorrelated (Spearman -0.141 between GRACE-FCast's lead-4 edge and the neighbor benefit), so the complementarity holds across basins as well as across leads. A genuinely hybrid product – filtered initial conditions, forcing-driven long leads, and a spatial correction – is the obvious system nobody has built; our splice is a descriptive sketch of its headroom, not that product.

5.3 The neighbor signal in context

Spatial context is increasingly assumed to help TWSA prediction (Li and Kusche, 2026; Arzoumanidis et al., 2026; Steidl and Zhu, 2025). Our results discipline that assumption more than they support it, and the two-sided answer is the contribution. On the discipline side, the linear result is a *null*: one correlation-selected neighbor adds a statistically indistinguishable third of a percent at lead 1 and nothing at longer leads (Sect. 4.4). Worse for the assumption, what little sits inside that null has a leakage signature – it is confined to basins whose mascon footprint is shared with land outside the basin and to leads 1–2, and it is *negative* in the 142 resolved, clean-footprint basins (Sect. 4.4). A graph attention network given identical information never beats a ridge, and two-hop neighborhoods never beat one-hop. Anyone building a basin graph for observed TWSA should expect these to be the default outcomes, and should stratify by footprint sharing before believing a positive one.

On the support side, the signal is not absent – it is inaccessible to the standard tools. Two nonlinear, two-stage constructions recover it: a neighbor-only residual MLP over a plain ridge at the GRACE-only information set and a neighbor-only correction stage over a deep own-state + ERA5 LSTM, each worth roughly one to two percent (Sects. 4.4 and 4.5). Both beat their full random-graph placebo families in every cell, so what is recovered is the neighbor's identity and not the extra stage. And the correction does not inherit the linear pocket's signature: it is positive in all 60 strata-by-lead cells, shows no statistically distinguishable difference across footprint-sharing terciles, and is worth $+0.89$ to $+2.00\,\%$ in precisely the resolved, clean-footprint stratum where the linear effect is negative. Along the size axis it does vary, in the direction that helps rather than hurts the reading: the smallest basins – the ones leakage would favour most – gain the least. That rules out the leading artifact explanation. It does not identify a mechanism, and we advance none: what the two-stage delivery extracts from a neighbor basin's history remains an open question, and calling it a teleconnection would be an overclaim. Steidl and Zhu (2025) forecast with a 10-nearest basin graph but on a reconstructed target, with no persistence-class baselines, no isolation of the graph's contribution, and no null models; our GAT null, control battery, and footprint-sharing stratification are the coherent companion

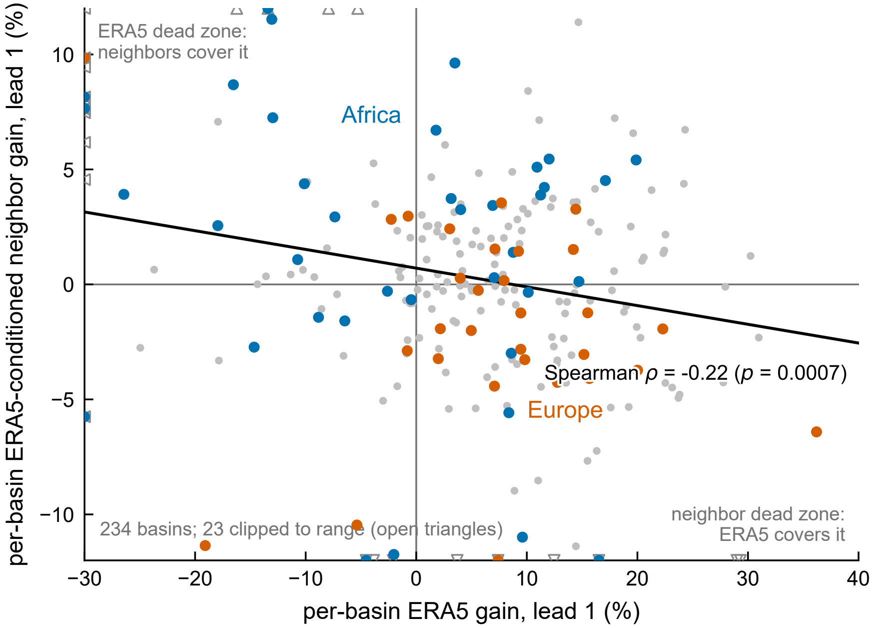

Figure 7. Do the two information sources cover each other's dead zones? Scatter of per-basin lead-1 ERA5 gain against per-basin lead-1 ERA5-conditioned neighbor gain for all 234 basins (gray), with the two continents the text contrasts highlighted: Africa (blue) and Europe (orange). The black line is a Theil–Sen fit; the rank correlation is -0.220 ($p = 0.0007$). Continental medians run from Africa ($+0.7\,\%$ ERA5, $+3.1\,\%$ neighbor) to Europe ($+7.6\,\%$ ERA5, $-1.9\,\%$ neighbor). Points outside the plotted range are pinned at the frame as open triangles.

measurement – the graph assumption, tested, yields a linear null with a footprint-sharing pocket, a small nonlinear signal that survives the leakage stratifier, and the finding that message passing is the wrong way to extract it.

A regional reading of where the neighbor channel pays would be valuable in a water-security context (Fig. 7), particularly the hypothesis that spatial TWSA information matters most where reanalysis forcing is least trustworthy. The corrected pipeline supports it from three directions. Per basin, ERA5 gain and conditioned neighbor gain are anti-correlated (Spearman -0.220, $p = 0.0007$, 234 basins). By continent, the two are mirror images: Africa has the smallest median ERA5 gain ($+0.7\%$) and the largest median neighbor gain ($+3.1\,\%$ conditioned; median per-basin skill $+1.6\,\%$ on the linear tier), while Europe shows the reverse ($+7.6\,\%$ ERA5, $-1.9\,\%$ neighbor). And pooled Africa sharpens rather than dilutes under conditioning: the unconditioned African neighbor effect is $+0.85\,\%$ ($p = 0.15$), unchanged by index conditioning, but rises to $+1.57\,\%$ ($p = 0.0035$) once all eleven ERA5 variables are aboard – spatial TWSA information is worth most precisely where the reanalysis has least to say. The GRACE-only residual-MLP result points the same way (Sect. 4.4).

5.4 Comparability of published numbers

No published study reports pooled deseasonalized basin RMSE over a global sample, so no external number is directly comparable to ours; we compare only against baselines recomputed under our protocol, and we flag two specific numbers a reader might juxtapose. Mo et al. (2025) report median grid-cell RMSE of 1.79/2.07/2.26 cm at 1–3 months; that figure is a median over grid cells of *full-signal* errors (seasonal cycle included) on a different window, three choices that each shrink it relative to our pooled deseasonalized basin metric, and only their code – not their predicted fields – is archived, so no matched-protocol comparison is possible. Li et al. (2026) report $\sim$1 cm domain-averaged deseasonalized RMSE for Africa; spatial averaging over the domain removes most variance, and their basin-level skill is categorical (ROC), not RMSE. We also note the FLDAS-Forecast hindcast fields are public (https://ldas.gsfc.nasa.gov/fldas/models/forecast); we did not add a third head-to-head because the product is regional (its Africa/Middle East domain covers $\sim$37 of our 234 basins), percentile/ROC-oriented, and ends in December 2020, overlapping only $\sim$19 months of our test targets. Its published conclusion – that initial-condition persistence dominates sub-seasonal TWS skill (Li et al., 2026) – is the qualitative claim our benchmark formalizes.

5.5 Limitations

1. **Stage-2 residuals are in-sample, and this is now measured rather than assumed.** The headline stacked correction is trained on the LSTM's training-set residuals. No test information reaches the MLP (the stage-1 network is shared), but because the LSTM overfits its training set the learned correction is plausibly attenuated. A variant refit on out-of-fold residuals bounds the concern, fitting stage 2 on three contiguous blocks of issue months held out of the stage-1 fit. The out-of-fold correction is *larger* at every lead ($+1.14$ to $+2.28\,\%$, against $+0.91$ to $+1.96\,\%$ in-sample; the direct contrast is $+0.20$ to $+0.34\,\%$, significant at all six leads) and beats 20 of 20 placebos in every cell. The in-sample protocol was therefore conservative, and the headline numbers we report understate the effect. The residual caveat that remains is the one this mechanism explains rather than inflates: it is why the own-state control learns a systematically wrong-direction correction (Sect. 4.5).

2. **The rank statistic is pinned at its floor.** With 20 placebo draws per cell the rank p cannot fall below $1/21$, which is why we report placebo standard deviations alongside it. Draws are redrawn independently per fold and lead, but seeds share a cell's draws, so the twelve lead–seed cells of Table 6 are six independent draw sets each scored twice.

3. **Controls not yet ported to the stacked system.** The $\geq$300 km exclusion and meteorology conditioning that bound the leakage and shared-climate confounds were run for the linear estimate, not for the stacked correction; the stacked evidence rests on the seed-matched placebo graphs alone. Indirect, and disclosed as such.

4. **Comparison asymmetries with GRACE-FCast.** Their training uses the Yin et al. (2023) reconstruction across the mission gap (hindsight); their seasonal/trend handling contributes $\sim$0.07–0.09 of the full-product edge at leads 4–6 (bounded by the non-seasonal variant, which still wins those leads); our lead-1 win partly reflects the Kalman filter being tuned to exactly this target definition; and the matched sample excludes fold 5 and seven island basins.

5. **Splice numbers are post hoc** (Sect. 4.3): splice points chosen on the evaluated sample, several variants, no multiplicity control.

6. **Sample and era.** One mascon product (CSR); test windows all lie in the post-2019 climate era, and any leave-one-fold-out selection, though causal in time, cannot escape that era; per-basin estimates rest on 83 lead-1 months.

7. **Seed budgets.** Two seeds for the LSTM stages, three for the residual MLP. Seed spread is comparable to the effects being measured – the LSTM sequence gain nearly halves from the better to the worse seed, the ERA5 residual MLP has a seed-2 outlier that flips sign at lead 1, and the LSTM neighbor increment at lead 2 flips sign between seeds. This is why every headline is a seed ensemble and no single-seed number is reported as a result.

8. **The filter's noise parameter is not identified as measurement noise.** 259 of 1170 basin–fold fits push r to its lower boundary and 31 basins sit there in all five folds, and the AR(1)-plus-white-noise decomposition absorbs unmodeled dynamics into r (Sect. 3.3); an $r = 0$ ablation and replication on JPL/GSFC mascons are the concrete checks.

9. **Post-selection inference.** All architectures were developed and compared on the same five test folds; the stacked system combines winners of those comparisons, so its results are exploratory (Sect. 4.5) and nominal p values understate total uncertainty. No untouched GRACE-FO period exists for confirmation within this record.

10. **Shared mascon footprints carry what pooled linear neighbor effect there is.** The footprint-sharing stratification (Sect. 4.4) is the primary sensitivity for the linear neighbor result: among basins GRACE resolves, it places the pooled estimate in the shared-footprint stratum and turns it negative elsewhere, with one small low-sharing cell ($+0.91\,\%$, $p = 0.036$, 14 basins) flagged as the exception.

11. **The footprint-sharing score is a lower bound, and the stratifications are not independent samples.** The score counts only foreign *land* – with land defined by CSR's official v02 land mask and tile assignment by the official mapping file (Sect. 2.1) – so the ocean fraction of a coastal tile is not counted as foreign and island and coastal basins score low; true footprint sharing is therefore at least as large as we measure. The linear 2×2 is lead 1 on the neighbor ridge only; the stacked stratification covers leads 1–6 but reuses the same test months as the headline, so it is a decomposition of the headline rather than an independent replication.

6 Conclusions

For basin-average, deseasonalized GRACE TWSA, the field's standard persistence benchmark is weaker than its own reference class: a three-parameter per-basin state-space filter applied to the same observations beats damped persistence by 5.0–8.8 % at leads 1–2 and the per-basin ridge – the best conventional model in our ladder at short leads – at five of six leads, at negligible cost. The margin is insensitive to the evaluation-protocol choices we varied (Sect. 4.1), so it is a property of the benchmark rather than of an implementation. We propose it as the default reference for this task, and we read the consequence constructively: benchmarked properly, short-lead TWSA forecasting is revealed to be a filtering problem, and the meaningful question becomes what information beats state-space filtering.

The answer has three parts. First, exogenous forcing: ERA5 predictors add about $5\,\%$ at lead 1, a twelve-month window of the same forcing extends the gain through lead 3 – and a linear reader of that window beats every sequence model we trained on it. Second, the crossover: a published system built entirely on exogenous forcing (Li and Kusche, 2026) overtakes our best system beyond lead 2, an empirical crossing consistent with the classic initial-conditions-versus-forcing partition, here located on satellite-observed TWSA. Third, cross-basin information, the two-sided part. At the linear tier it is a null, and what little sits inside that null looks like an instrument artifact: the estimate concentrates in basins whose mascon footprints are shared with outside land and turns negative where footprints are clean. Representation is what changes the answer. Handed over as raw 12-month history, the same neighbor signal yields nothing, whether as an encoder input channel or as a dedicated correction stage; as a single state propagated to the forecast target time by the filter, it contributes roughly 1–$2\,\%$ at every lead. Two controls carry that claim, and neither is the leaderboard: the degree-matched placebo family, which makes it a statement about the neighbor's information rather than about the extra stage, and the footprint-sharing stratification, in which the correction is positive in every stratum at every lead and is, if anything, weakest in the basins an artifact would favour. The artifact that explains the linear pocket therefore does not explain the correction; what does explain it remains unidentified.

Short leads are a filtering problem; long leads are a forcing problem; the spatial signal is small, invisible to linear methods, and usable only in propagated-state form. The obvious next system combines all three, and the obvious next practice is to benchmark it – and everything else in this literature – against persistence of the filtered state rather than persistence of the raw observation.

Code and data availability. CSR GRACE/GRACE-FO mascon solutions are available from the Center for Space Research (https://www2.csr.utexas.edu/grace/). ERA5-Land data are available from the Copernicus Climate Data Store (https://cds.climate.copernicus.eu/) (Muñoz-Sabater et al., 2021). Climate indices are available from NOAA PSL (https://psl.noaa.gov/data/climateindices/). The GRACE-FCast hindcast is available from PANGAEA (Li, 2025). Analysis code, derived basin tables, and all per-prediction result files sufficient to reproduce every number in this paper will be archived on Zenodo upon acceptance; Appendix A maps the model names used in this paper to the identifiers in those files.

Appendix A: Model names and archived identifiers

The archived result files identify each model by the internal identifier used in the analysis code. Table A1 maps the model names of Table 1 to those identifiers; the identifiers appear nowhere else in the paper. Seed suffixes (`_s0`, `_s1`, `_s2`) denote single training seeds and `_ens` the seed ensemble. The archive's manifest (`README`) maps each table and figure to the result file it is drawn from.

Table A1. Model names used in the paper and the corresponding identifiers in the archived result files.

Paper name	Identifier(s)
Climatology	`climatology_zero`
Persistence	`persistence`
Damped persistence	`damped_persistence_rho`, `damped_persistence_reg`
Pooled ridge (and climate-index variant)	`ridge_own_lags`, `ridge_own_plus_indices`
Per-basin ridge	`ridge_own_perbasin`
Kalman forecast	`kalman_ar1`
Ridge on filtered states	`kalman_own_ridge`
Neighbor ridge (pre-specified contrast)	`kalman_corr_top1`
Neighbor ridge, distance-restricted	`kalman_corr_min300_top1`, `kalman_corr_min500_top1`, `kalman_corr_min1000_top1`
Neighbor ridge, alternative graphs	`kalman_geo_top1`, `kalman_geo_top2`, `kalman_geo_top3`, `kalman_corr_top2`, `kalman_corr_top3`
Own-basin ridge / ERA5 ridge	`ridge_own`, `ridge_own_era5`
Neighbor ridge (forcing experiments)	`ridge_corr_top1`, `ridge_corr_top1_era5`
GBM head	`gbm_own_era5`, `gbm_corr_top1_era5`
MLP head	`mlp_own_era5_s0`
Residual MLP	`resmlp_corr_top1`, `resmlp_corr_top1_era5`, `resmlp_own_era5`
LSTM (and neighbor-channel variant)	`lstm_own`, `lstm_own_era5`, `lstm_corr_top1_era5`
Stacked system	`lstmres_corr_top1`
Own-state control	`lstmres_own`
GAT	`gnn_corr_top1`, `gnn_corr_top2_era5`, and variants
GRACE-FCast (full / non-seasonal)	`li_lstm_full`, `li_lstm_nonseas`

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